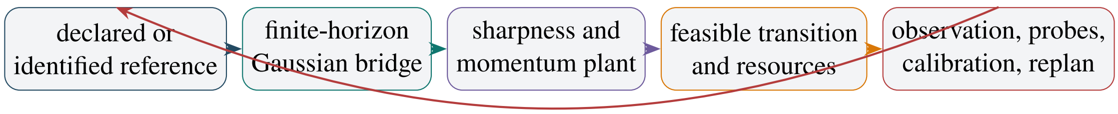
Bridge Learning

Reference-Consistent Covariance Steering
with Dynamic Sharpness and Momentum

From OneCycle and Schrodinger-bridge geometry
to identified stochastic plants, persistent excitation, and safe control software



Revised technical research report accompanying `bridgelearn` v0.3.0

July 2026

Research status. This document develops a falsifiable control framework rather than a claim that neural-network training is literally a Schrodinger bridge. Brownian bridge theory explains why endpoint constraints can generate OneCycle-like covariance humps. The operational algorithm instead uses a reference-consistent finite-horizon linear-Gaussian bridge, then realizes its transitions through identified local dynamics. Version 0.3 promotes sharpness to a slow controlled state, treats momentum through companion-form dynamics, uses order-2 antithetic excitation when cold identification fails, and makes non-normal stability, compute, and calibration explicit.

Abstract

OneCycle learning-rate policies are powerful open-loop heuristics: they commit to a horizon, warm a scalar learning rate toward a prescribed maximum, often cycle momentum inversely, and then anneal toward a small floor. Brownian Schrodinger bridges explain an important geometric part of this behavior. Their covariance observables are quadratic in intrinsic time, and endpoint covariance asymmetry determines whether the dispersion maximum occurs early, late, at the midpoint, or at a boundary. This provides a mathematical explanation for the existence and skew of a one-cycle-like hump, but marginal spread is not itself a learning rate or a complete stochastic dynamics.

Bridge Learning replaces the fixed curve with a receding-horizon covariance-steering problem. For a declared or identified linear-Gaussian reference, the finite-horizon bridge supplies a coherent sequence of marginal covariances, adjacent cross-covariances, contractions, and innovations. The same reference governs path and coupling, avoiding the inconsistency of tracking Brownian marginals through transitions optimized relative to a different, contracting process. A controller then projects each requested transition onto the learning system's available controls: step size, damping, momentum, effective batch, gradient accumulation, rollout resource, or explicit process noise. A feasibility clock and horizon governor expose additional physical updates and compute rather than hiding them in a nominal schedule.

The newest extension addresses two first-order obstacles to realistic deployment. Near the edge of stability, curvature is promoted from a static response curve to a slow identified sharpness state with progressive sharpening, threshold restoration, robust prediction funnels, model adequacy, and jump detection. Momentum channels are represented by companion-form AR(2) dynamics. Known commands permit gray-box reconstruction of the local gradient sample; exact antithetic probe pairs supply persistent excitation in the weak order-2 contrast direction; Jury conditions define the stability triangle; a Lyapunov-adapted covariance norm exposes non-normal transient amplification; and fixed momentum produces an explicit cooling-reachability floor.

The accompanying `bridgelearn` v0.3.0 library implements these ideas as a backend-neutral control layer. It supports persistent channels for weights, actors, critics, and recurrent parameters, and episodic channels for predictive-coding activity, fixed-point residuals, or itinerant control states. The distribution includes dynamic sharpness, momentum identification, antithetic probing, companion stability, compute-priced discrete resources, robust width scales, one-step calibration, shadow mode, and checkpointable failure states. Thirty-eight tests and six small sandbox examples validate the declared local equations and software behavior. They do not establish neural-task superiority; task utility, compute, sharpness, basin or trajectory evidence, and calibration remain separate scientific outcomes.

Executive summary

Central claim

Bridge Learning is not a bridge-shaped learning-rate curve. It is a receding-horizon realization of a finite-horizon stochastic transition law. The operational path, cross-time coupling, and local reference come from one plant model; identified sharpness and momentum determine reachability; actuator limits, uncertainty, physical time, and compute determine how quickly a learning system may follow the path.

1. **Brownian bridge theory explains shape.** Every covariance projection of a Brownian Schrodinger bridge is quadratic in normalized time. Endpoint covariance asymmetry determines the direction of peak skew. Gaussian endpoints yield closed-form peak location, height, floor, and boundary-pinning phase transitions.

2. **The operational bridge is reference-consistent.** The linear-Gaussian path and receding-horizon planner derive marginal widths, adjacent cross-covariances, and transitions from one declared scalar or diagonal reference. Fixed or slowly filtered references avoid last-command hysteresis.

3. **The controller steers transitions, not just marginals.** Cross-time covariance determines memory and contraction; the Schur-complement innovation determines new stochastic variation. This separates movement from temperature and resolves the two-root ambiguity of one-step marginal inversion.

4. **Sharpness is a state.** A static fit $h(\eta)$ cannot represent progressive sharpening, lag, or attraction toward a moving stability threshold. Version 0.3 identifies a slow scalar sharpness plant, propagates a lower/center/upper funnel, and plans against predicted future sharpness.

5. **Momentum changes reachability.** Heavy-ball dynamics are companion-form AR(2). Jury conditions give the edge $\eta h < 2(1 + \mu)$; underdamped poles have radius $\sqrt{\mu}$; non-normality can transiently amplify covariance error even when every pole is stable.

6. **Cold identification requires temporal excitation.** White-noise amplitude does not repair a starved order-2 regressor direction. Exact $+\delta, -\delta$ antithetic pairs place information near the Nyquist frequency, directly exciting the $c_1 - c_2$ contrast while remaining pairwise mean-zero.

7. **Uncertainty is asymmetric.** Stability is checked at the upper sharpness bound; progress is promised only at the lower bound. A wide funnel, structural mismatch, exhausted probe budget, or catapult jump is a named failure state.

8. **Compute and calibration are first-class outcomes.** Effective batch and accumulation are discrete and costly. The controller prices resource use, applies hysteresis, reports progress per compute, and produces predicted-versus-realized one-step calibration artifacts.

9. **The architecture is backend-neutral.** The core controls transitions rather than gradients.

Backpropagation, predictive coding, fixed-point inference, reinforcement learning, and CAROM differ through adapters, observers, and actuators rather than through separate mathematical cores.

10. **Covariance tracking is not a task theorem.** Matched widths can hide different lag correlations, anisotropic innovation, sharpness, implicit regularization, basin occupation, and trajectory use. Mechanistic claims require richer measurements.

Revision history and scientific boundary

The path/coupling/actuator/clock factorization survived two rounds of critique. Version 0.3 unified the reference process, added compute and calibration, and made observer adequacy explicit. Version 0.3 replaces the static curvature ontology with a dynamic sharpness plant and promotes momentum identification, persistent excitation, companion stability, and reachability to core machinery. The remaining limitations are explicit: the bridge path is still scalar or diagonal; companion actuation is not yet a full augmented-state Schrodinger bridge; the sharpness law is phenomenological; dual-control probe allocation is heuristic; and task-level benefits remain to be demonstrated.

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Chapter 1

Motivation, scope, and scientific status

1.1 Why revisit OneCycle learning?

Learning-rate schedules are among the most consequential and least mechanistically transparent components of modern training pipelines. Their practical role is broader than changing the distance moved along a descent direction. Through minibatch noise, momentum, preconditioning, clipping, weight decay, and nonlinear curvature, a learning-rate change can alter local stability, stochastic exploration, basin escape, implicit regularization, and the timescale on which different parameter groups reorganize.

The one-cycle policy was introduced in the context of cyclical learning rates and super-convergence [14, 15, 16]. In its common two-phase implementation, a learning rate is raised from a small initial value to a large maximum and then annealed to a very small final value. The PyTorch implementation requires a total number of updates, uses a default warm-up fraction of 0.3, and normally cycles momentum inversely to the learning rate [12]. These choices are effective empirical defaults, but they also expose a conceptual gap: the schedule is a prescribed function of batch count, not a response to the geometry or stochastic state of the particular run.

This gap became concrete in the CAROM experiments that motivated the present investigation. In a 12,000-update frozen-GPT-2 CAROM run, frozen endpoint accuracy peaked at update 3,000 and then collapsed as the stretched `OneCycleLR` schedule approached its maximum. Later annealing did not recover the earlier solution. Meanwhile, dependency-edge accuracy remained nearly unchanged or improved, whereas itinerary structure, trajectory coverage, the natural-minus-shuffled causal gap, and downstream execution performance deteriorated [4]. The result is not proof of schedule-induced failure: it is a single seed, and schedule duration changed together with training duration. It is nevertheless a sharp illustration of why one global, open-loop scalar may be poorly matched to a modular recurrent architecture.

The same need appears in other learning paradigms. Predictive-coding networks contain an inner inference process over activities or errors and an outer learning process over weights. Deep-equilibrium models turn depth into a fixed-point solve whose iteration count depends on contraction and residual tolerance [18, 1]. Reinforcement learning separates actor, critic, representation, target, and entropy-temperature updates, all under nonstationary data. CAROM adds scheduled, fixed-point, and itinerant recurrent semantics, each with distinct dynamical controls [3, 5]. A scheduler tied only to global update count cannot express these multiple channels and timescales.

1.2 The bridge-learning question

The guiding question is not

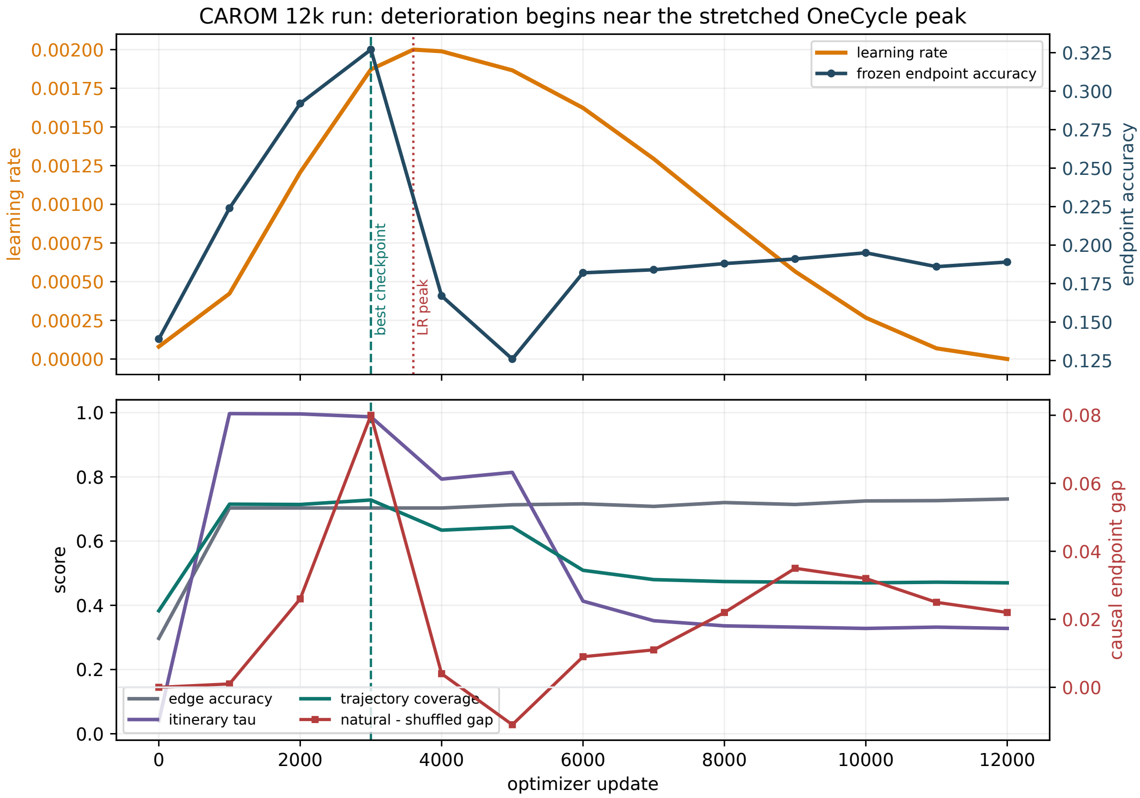


Figure 1.1: CAROM motivation for closed-loop, blockwise control. The upper panel reconstructs the 12,000-update learning-rate trajectory and overlays frozen endpoint accuracy. The lower panel shows that edge prediction remained stable while itinerary and trajectory evidence deteriorated. The vertical lines mark the best checkpoint and the approximate learning-rate peak. The experiment is a single-seed, compound change in duration and schedule, so the timing is circumstantial rather than causal proof. Data are transcribed from the attached CAROM 12k report [4].

> What analytic curve should replace the OneCycle cosine?

but rather

> Can boundary conditions define a desired stochastic evolution, and can local measurements determine how an actual learning algorithm should realize that evolution?

Schrodinger bridges provide the first half of this answer. A bridge is a path-space inference problem: among stochastic evolutions compatible with specified endpoint distributions, choose the one closest in relative entropy to a reference process [10, 6]. This has the same fixed-horizon, pinned-endpoint structure that OneCycle implicitly assumes. A scalar Gaussian surrogate indeed produces a one-cycle-like dispersion hump and an endpoint-asymmetry law for its peak [13].

The second half requires a correction. Bridge marginal dispersion is not the same object as learning rate or local diffusion. A bridge covariance path does not determine the cross-time coupling of consecutive states. Different learning rates may produce the same next marginal variance while having radically different memory, contraction, and innovation. Therefore, the mathematically natural object to control is not a scalar schedule but a local stochastic transition.

Bridge Learning in one sentence

Choose a boundary-conditioned covariance path, complete each pair of consecutive marginals with a preferred contraction-and-innovation transition, and project that transition onto the feasible controls of the learning system.

1.3 Scope of this report

This report has five goals.

First, it retains the exact Brownian covariance theory as background for understanding OneCycle-like shapes. That theory is useful and general, but it is explicitly separated from the operational controller.

Second, it derives the exact finite-horizon bridge for a scalar or diagonal linear-Gaussian reference. This supplies marginal covariances and cross-time transitions from one coherent reference process.

Third, it develops a constrained adaptive-control formulation with declared reference policies, actuator projection, horizon governance, confidence-gated probing, compute-priced shared resources, and calibrated one-step prediction.

Fourth, it explains the critique-driven `bridgelearn` v0.3.0 architecture and its safe deployment rules for momentum, centering, heavy-tailed observables, and curvature response.

Fifth, it relates the framework to CAROM, predictive coding, fixed-point inference, backpropagation, and reinforcement learning and defines a falsifiable, compute-matched experimental program.

The report does not claim that neural-network optimization obeys a global linear-Gaussian model. The local model is a controlled approximation. Multimodal basin occupation, representation learning, optimizer memory, heavy-tailed noise, data nonstationarity, and the endogeneity of curvature can all invalidate a covariance-only description. The intended scientific workflow is therefore shadow observation, adequacy testing, one-step calibration, conservative active control, and explicit task-level ablation.

Table 1.1: Assessment of the major critique themes and their disposition in v0.3.

Theme	Assessment	Revision
Brownian path versus optimizer reference	Accepted	Exact linear-Gaussian reference bridge is now the default; Brownian mode is legacy/theory.
Last-command hysteresis	Accepted	Fixed, callback, and slowly filtered commanded references are first-class.
Cold adaptive observer	Accepted	Confidence gates and bounded innovation probes; stability claims are conditional.
Curvature endogeneity	Accepted locally	First-order response model plus trust-region and edge-of-stability diagnostics.
Momentum misspecification	Accepted	AR(2) adequacy is required for momentum/Adam active use; low adequacy blocks deployment.
Centering and heavy tails	Accepted	Explicit centering observer and robust scale estimators.
Covariance sufficiency	Rejected as a claim	Width is one controlled observable, not a sufficient statistic for generalization.
Compute fairness and calibration	Accepted	Compute-priced quantized resources, compute-matched example, and first-class calibration artifacts.

1.4 Notation

The main notation is summarized in Table 1.2. Scalars are used for the block-scalar implementation; bold or matrix notation is used for the conceptual generalization.

Table 1.2: Main notation.

Symbol	Meaning
$s \in [0, 1]$	Intrinsic bridge progress, not necessarily physical iteration fraction.
q	Total Brownian quadratic variation in the explanatory Brownian model.
V_0, V_T, V_k	Initial, terminal, and intermediate scalar active variance.
Σ_k	Matrix covariance of a centered local learning state.
A_k	Linear contraction or memory map between consecutive centered states.
R_k	Covariance of new stochastic innovation.
C_k	Cross-time covariance $\mathrm{Cov}(z_{k+1}, z_k)$.
η_k	Learning rate, relaxation step, or another movement actuator.
h_k	Effective positive local curvature or contraction coefficient.
ν_k	Unit-resource stochastic update-noise scale.
m_k	Effective batch, trajectory batch, accumulation, or other noise-reducing resource.
a_0, r_0	Contraction and innovation of the declared local reference process.
N_k	Remaining physical horizon used by the reference-consistent planner.
ρ_0, ρ_T	Endpoint distributions in a Schrodinger bridge.
R or R	Reference path measure; context distinguishes it from innovation covariance.

Chapter 2

OneCycle as an open-loop temperature policy

2.1 The two-phase schedule

Let $t \in [0, 1]$ denote normalized physical training time and let $p \in (0, 1)$ be the warm-up fraction. A representative two-phase cosine OneCycle schedule is

$$\eta(t) = \begin{cases} \eta_0 + \dfrac{\eta_{\max} - \eta_0}{2} \left[1 - \cos\left(\dfrac{\pi t}{p}\right)\right], & 0 \le t \le p, \\ \eta_{\min} + \dfrac{\eta_{\max} - \eta_{\min}}{2} \left[1 + \cos\left(\dfrac{\pi(t-p)}{1-p}\right)\right], & p < t \le 1. \end{cases} \tag{2.1}$$

In PyTorch, the default parameters imply

$$p = 0.3, \qquad \eta_0 = \frac{\eta_{\max}}{\texttt{div\_factor}}, \qquad \eta_{\min} = \frac{\eta_0}{\texttt{final\_div\_factor}}, \tag{2.2}$$

with default divisors 25 and 10^4, respectively [12]. Momentum is normally cycled inversely to η.

The policy has a useful qualitative interpretation. The rising phase permits larger moves and stronger stochastic exploration. The long descending phase suppresses instability and permits local settling. Large learning rates can also provide regularization, which is a key part of the super-convergence argument [15, 16]. These are real benefits, and Bridge Learning should be viewed as a generalization rather than a dismissal of OneCycle.

2.2 Hidden assumptions

Equation (2.1) contains several implicit modeling assumptions.

2.2.1 The clock is the update count

The scheduler assumes that the same fraction of batch updates has the same dynamical meaning across runs. Yet changing batch size, data curriculum, gradient noise, curvature, optimizer state, or recurrent inference budget changes the amount of stochastic and geometric progress per update. Stretching the total horizon also stretches the location and duration of the high-learning-rate regime. This is exactly what occurred in the CAROM 12k experiment: at update 3,000 the 12k run was still near its rising peak, whereas the earlier 4k schedule had already annealed substantially [4].

2.2.2 One scalar controls all parameter groups

Modern optimizers support parameter groups, but a single OneCycle curve usually imposes one normalized temporal shape on all of them. Different blocks may have different curvatures, noise scales, terminal objectives, and phase transitions. In CAROM, the edge head remained accurate while downstream routing and execution degraded; a blockwise controller could cool or freeze the solved component while continuing to adapt the unsolved components.

2.2.3 Learning rate is asked to control two effects

A learning rate changes deterministic contraction and stochastic innovation simultaneously. In a local scalar noisy quadratic,

$$x_{k+1} = (1 - \eta_k h_k)x_k + \eta_k \xi_k, \qquad \mathrm{Var}(\xi_k) = \frac{\nu_k}{m_k}, \tag{2.3}$$

so

$$V_{k+1} = (1 - \eta_k h_k)^2 V_k + \eta_k^2 \frac{\nu_k}{m_k}. \tag{2.4}$$

Increasing η_k may contract the old state more strongly while also injecting more new variance. There is no universal monotone map from marginal variance to learning rate.

2.2.4 The endpoint objective is implicit

The final near-zero learning rate encodes an implicit low-temperature terminal condition. That may be appropriate for a sharply converged point estimate, but not necessarily for stochastic weight averaging, continued pretraining, posterior exploration, or a recurrent system whose noise controls computational dwell time. Stochastic weight averaging, for example, deliberately samples a region using a nonzero or cyclical learning rate before averaging [8]. A higher-entropy terminal objective should not automatically inherit the same early peak and near-zero floor.

2.3 Why a boundary-value formulation is attractive

OneCycle already commits to a total horizon and a terminal state. What it lacks is a principle that maps endpoint intent into the path shape and a feedback mechanism that maps the desired path into feasible controls. Schrodinger bridges provide a canonical answer to the first question:

> Among stochastic path measures satisfying the endpoint constraints, choose the one closest in relative entropy to a reference dynamics.

This has three attractive consequences.

1. The path is derived from initial and terminal distributions and a reference stochastic budget.
2. The solution includes temporal coupling, not just pointwise marginal widths.
3. The same mathematics connects path-space inference, stochastic optimal control, entropic optimal transport, and covariance steering [10, 6, 7].

The first scalar experiments showed that a Gaussian bridge can reproduce the OneCycle hump and its early-peak asymmetry [13]. Figure 2.1 displays that comparison. The rest of this report explains what is exact in that observation and what must be added before it becomes a learning algorithm.

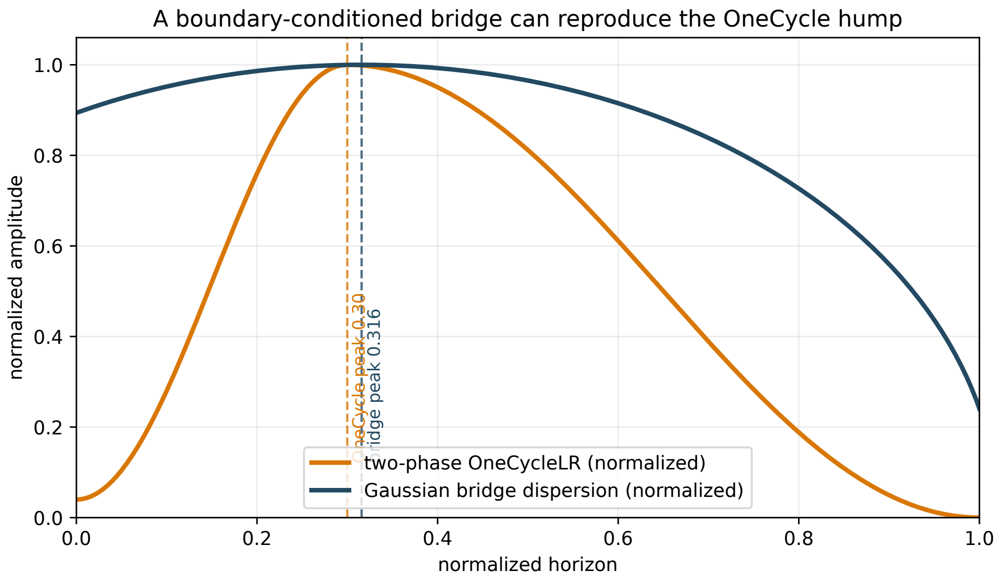


Figure 2.1: A normalized two-phase OneCycle learning-rate curve and the normalized standard deviation of a scalar Gaussian Brownian Schrodinger bridge. The endpoint variances and diffusion budget are chosen so that the bridge peak occurs at $s \simeq 0.316$, close to the default OneCycle peak at 0.30. This is a shape comparison, not an identification of learning rate with marginal dispersion.

Chapter 3

Brownian Schrodinger bridges as an explanatory baseline

Role of this chapter in the revised framework

The results in this chapter remain mathematically valid and explain why endpoint conditions can generate a one-cycle-like covariance hump. They are not the default operational controller. Beginning with the next chapter, the path and transitions are generated from the same identified linear-Gaussian reference process.

3.1 Path-space relative-entropy projection

Let Ω be a path space over $[0, T]$, let R be a reference path measure, and let P_0 and P_T denote the initial and terminal marginals of a candidate path measure P. The Schrodinger bridge problem is

$$P^\star = \arg\min_{P:\, P_0=\rho_0,\, P_T=\rho_T} \mathrm{KL}(P \,\|\, \mathsf{R}). \tag{3.1}$$

This formulation dates to Schrodinger's question about the most likely evolution of a cloud of diffusing particles observed with atypical endpoint distributions. Modern treatments connect it to reciprocal processes, entropic optimal transport, stochastic control, and large deviations [10, 6].

For a diffusion reference

$$\mathrm{d}X_t = b_{\mathsf{R}}(t, X_t)\,\mathrm{d}t + \sqrt{\varepsilon}\,\mathrm{d}W_t, \tag{3.2}$$

the bridge may be represented as a controlled diffusion with the same local diffusion coefficient and an altered drift,

$$\mathrm{d}X_t = \big[b_{\mathsf{R}}(t, X_t) + u_t(X_t)\big]\mathrm{d}t + \sqrt{\varepsilon}\,\mathrm{d}W_t, \tag{3.3}$$

where the control minimizes an expected quadratic energy subject to the endpoint laws. The exact normalization depends on the reference convention, but the essential point is invariant: finite path-space relative entropy changes the drift while preserving the diffusion's quadratic variation.

This observation will later prevent a common misinterpretation. The bridge's marginal standard deviation is accumulated state dispersion. It is not the local diffusion coefficient and therefore is not directly a learning rate.

3.2 Endpoint coupling and the reciprocal property

A Schrodinger bridge can be decomposed into two parts:

1. an optimal endpoint coupling $\pi^\star(\mathrm{d}x_0, \mathrm{d}x_T)$; and
2. the reference process conditioned on those endpoints.

More precisely, the bridge lies in the same reciprocal class as the reference: conditional on (X_0, X_T), its path law is the reference bridge. For a Brownian reference, this means that the interpolation between fixed endpoints is an ordinary Brownian bridge. This single fact yields the covariance theorem below.

Let

$$\mathrm{d}X_t = \sqrt{\varepsilon}\,\mathrm{d}W_t, \qquad 0 \le t \le T, \tag{3.4}$$

and define

$$s = \frac{t}{T}, \qquad q = \varepsilon T. \tag{3.5}$$

Conditional on the endpoints,

$$X_t = (1-s)X_0 + sX_T + \sqrt{q\,s(1-s)}\,Z, \tag{3.6}$$

where $Z \sim \mathcal{N}(0, I)$ is independent of the coupled endpoints.

Theorem 3.1 (Quadratic covariance law)**.** *Let $P^\star$ be a Brownian Schrodinger bridge between arbitrary endpoint distributions with finite second moments. Write*

$$\Sigma_0 = \mathrm{Cov}(X_0), \quad \Sigma_T = \mathrm{Cov}(X_T), \quad \Gamma = \mathrm{Cov}(X_0, X_T) \tag{3.7}$$

under the bridge endpoint coupling. Then the marginal covariance at intrinsic time $s = t/T$ is

$$\boxed{\Sigma_s = (1-s)^2\Sigma_0 + s^2\Sigma_T + s(1-s)\big(\Gamma + \Gamma^\top + qI\big).} \tag{3.8}$$

Thus every fixed linear scalarization of covariance is a quadratic function of intrinsic bridge time.

Proof. Apply the law of total covariance to Equation (3.6). Conditional covariance contributes $q\,s(1-s)I$. The covariance of the conditional mean $(1-s)X_0 + sX_T$ contributes the remaining endpoint and cross-covariance terms. No Gaussian assumption is required. □

For a fixed direction u or positive semidefinite metric M, define

$$v_u(s) = \mathrm{Var}(u^\top X_s), \tag{3.9}$$

$$v_M(s) = \mathrm{tr}(M\Sigma_s). \tag{3.10}$$

Both are quadratic. Therefore a Brownian bridge covariance observable can have at most one interior maximum. This is much more general than the original scalar Gaussian numerical surrogate.

3.3 Peak skew from endpoint covariance asymmetry

For a scalar direction, write

$$A = \mathrm{Var}(X_0), \qquad B = \mathrm{Var}(X_T), \qquad K = 2\,\mathrm{Cov}(X_0, X_T) + q. \tag{3.11}$$

Then

$$v(s) = A(1-s)^2 + Bs^2 + Ks(1-s). \tag{3.12}$$

If an interior maximum exists, it occurs at

$$s_\star = \frac{1}{2} + \frac{B-A}{2(K-A-B)}. \tag{3.13}$$

The interior-hump condition is

$$K > 2\max(A, B). \tag{3.14}$$

Corollary 3.2 (Endpoint-asymmetry peak law)**.** *Whenever the scalar bridge variance has an interior maximum,*

$$\operatorname{sign}\left(s_\star - \frac{1}{2}\right) = \operatorname{sign}(B - A). \tag{3.15}$$

The peak is early if terminal variance is smaller than initial variance, centered if they are equal, and late if terminal variance is larger.

The endpoint coupling changes whether a hump exists and how far the peak moves, but not the direction of the skew. Time reversal gives $s_\star(A, B) = 1 - s_\star(B, A)$.

This result sharpens the informal phrase "entropy asymmetry." For one-dimensional Gaussians, entropy order and variance order coincide. For general non-Gaussian distributions, they need not. The exact theorem is about covariance asymmetry and endpoint cross-covariance; entropy is a special Gaussian parametrization.

3.4 Closed form for Gaussian endpoints

Suppose

$$X_0 \sim \mathcal{N}(a, A), \qquad X_T \sim \mathcal{N}(b, B). \tag{3.16}$$

The optimal endpoint coupling is Gaussian. With the Brownian convention above, its scalar cross-covariance is

$$C_q = \frac{\sqrt{q^2 + 4AB} - q}{2}, \tag{3.17}$$

consistent with closed-form Gaussian entropic transport and Gaussian bridge results [9, 2]. Since $2C_q + q = \sqrt{q^2 + 4AB}$, the variance path becomes

$$\boxed{V(s) = A(1-s)^2 + Bs^2 + S\,s(1-s), \qquad S = \sqrt{q^2 + 4AB}.} \tag{3.18}$$

This is the formula implemented by `GaussianBridgePath` in `bridgelearn`.

Let

$$D = S - A - B. \tag{3.19}$$

In the interior-hump regime $S > 2\max(A, B)$,

$$\boxed{s_\star = \frac{S - 2A}{2D} = \frac{1}{2} + \frac{B - A}{2D}.} \tag{3.20}$$

The peak height simplifies to

$$\boxed{V_{\max} = \frac{q^2}{4D}.} \tag{3.21}$$

The terminal standard-deviation floor relative to the peak is therefore

$$\frac{\sqrt{B}}{\sqrt{V_{\max}}} = \frac{2\sqrt{BD}}{q}. \tag{3.22}$$

3.4.1 Boundary-pinning phase transitions

The quadratic path has three possible topologies.

If $B > A$, the interior peak reaches the terminal boundary when $S = 2B$, giving

$$B_+ = \frac{A + \sqrt{A^2 + q^2}}{2}. \tag{3.23}$$

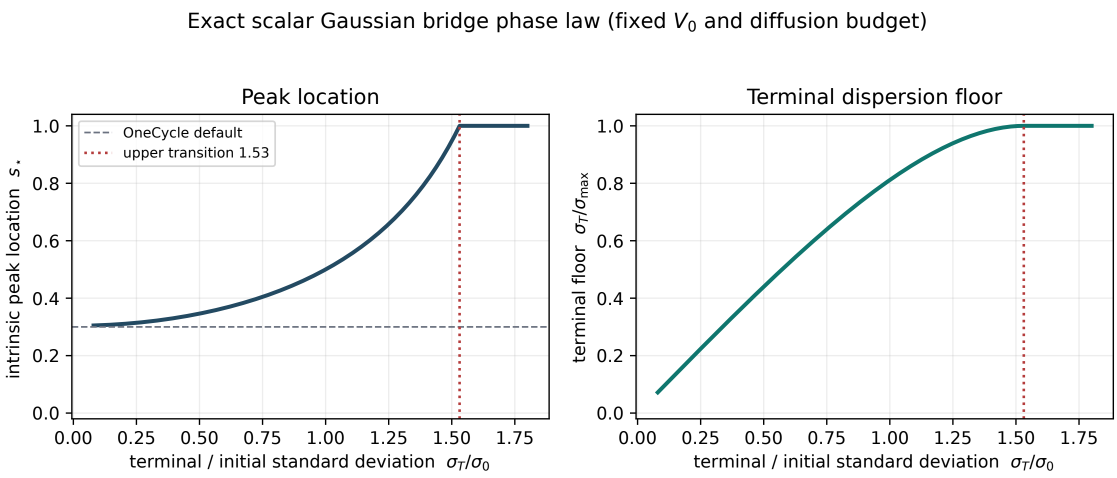


Figure 3.1: Exact scalar Gaussian bridge phase law at fixed initial variance and diffusion budget. As terminal width increases, the peak moves monotonically later and the terminal floor rises. At the upper transition, the interior hump disappears and the path becomes monotone warming. The plotted parameter values match the scale of the original scalar surrogate.

For $B \geq B_+$, variance rises monotonically and the peak is at $s = 1$.

If $A > B$ and the lower threshold is positive, the peak reaches the initial boundary when $S = 2A$, giving

$$B_- = A - \frac{q^2}{4A}. \tag{3.24}$$

For $B \leq B_-$, the path cools from the start and no warm-up is required.

The phase diagram is therefore

$$\text{cooling only} \quad \longrightarrow \quad \text{interior explore–anneal hump} \quad \longrightarrow \quad \text{warming only}. \tag{3.25}$$

3.5 Scale-free and scale-dependent quantities

Write

$$r = \frac{\sqrt{B}}{\sqrt{A}}, \qquad \lambda = \frac{q}{A}. \tag{3.26}$$

Then

$$s_\star(r, \lambda) = \frac{1}{2} + \frac{r^2 - 1}{2\left(\sqrt{\lambda^2 + 4r^2} - 1 - r^2\right)} \tag{3.27}$$

inside the hump regime. The law is invariant under common scaling

$$(A, B, q) \mapsto (cA, cB, cq), \tag{3.28}$$

but it is not parameter-free. It depends on both the endpoint width ratio r and the normalized diffusion budget λ. A sweep that fixes A and q may look one-dimensional, but the general family is two-dimensional.

3.6 The crucial obstruction: marginal spread is not local temperature

The bridge result explains a shape, but it does not directly define a learning rate.

For the Brownian reference, every finite-relative-entropy bridge has the same pathwise quadratic variation as the reference. The bridge changes drift, not diffusion coefficient. Yet its marginal variance changes substantially because drift, endpoint conditioning, and accumulated noise redistribute state uncertainty.

In stochastic optimization, two related quantities are also easily confused. If an update uses learning rate η and minibatch gradient-noise covariance Q/m, then the covariance injected per discrete update is approximately

$$\eta^2 \frac{Q}{m}. \tag{3.29}$$

If continuous optimization time is rescaled by $\mathrm{d}t \simeq \eta$, the corresponding diffusion rate is of order

$$\eta \frac{Q}{m}. \tag{3.30}$$

The frequently cited SGD noise scale η/m is therefore a rate in an optimizer-time convention, not a marginal parameter variance [17, 11].

3.6.1 Intrinsic-time non-identifiability

Let a time-inhomogeneous reference have diffusion rate $a(t)$,

$$\mathrm{d}X_t = \sqrt{a(t)}\,\mathrm{d}W_t. \tag{3.31}$$

Define cumulative quadratic variation

$$Q_t = \int_0^t a(u)\,\mathrm{d}u, \qquad r(t) = \frac{Q_t}{Q_T}. \tag{3.32}$$

The Brownian covariance formula holds with s replaced by $r(t)$. Endpoint marginals and total diffusion determine the path in intrinsic diffusion time, but not the physical clock $t \mapsto r(t)$. Any admissible time warp can move the peak in iteration time.

This motivates one of the defining features of Bridge Learning: the bridge path is specified in intrinsic progress, while a feasibility-aware clock determines how physical updates traverse that path.

Interpretation boundary

The exact bridge theory supports the statement "boundary conditions determine a covariance path in intrinsic time." It does not, by itself, support the statement "boundary conditions uniquely determine the learning-rate schedule." The latter requires a constitutive model relating optimizer controls to contraction, stochastic innovation, and physical time.

Chapter 4

Reference-consistent Gaussian bridges

4.1 Why one reference process must govern both path and transition

A Schrodinger bridge is defined relative to a reference path measure. It is therefore internally inconsistent to generate target marginals from a Brownian reference and then choose each local transition by relative entropy to a contracting optimizer reference. The composite trajectory is generally the bridge of neither reference. The discrepancy is largest exactly where a restoring force strongly alters the covariance evolution.

The operational revision begins with a declared or identified local reference

$$X_{k+1} = a_0 X_k + \sqrt{r_0}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0,1), \tag{4.1}$$

for each scalar block. In the diagonal case, all formulas below apply componentwise. The reference may represent a conservative baseline learning rate and batch, a damped predictive-coding relaxation, a critic update at declared rollout statistics, or a recurrent integration step with declared noise.

Given current variance V_0, terminal variance V_N, and remaining horizon N, the finite-horizon Schrodinger problem relative to Equation (4.1) determines one endpoint coupling and one reciprocal path law. Marginals and cross-time transitions are then different views of the same solution.

4.2 Reference moments and endpoint coupling

Define the k-step state-transition factor and controllability Gramian

$$\phi_k = a_0^k, \tag{4.2}$$

$$Q_k = r_0 \sum_{j=0}^{k-1} a_0^{2j}, \tag{4.3}$$

with $\phi_0 = 1$ and $Q_0 = 0$. Let $\Phi = \phi_N$ and $Q = Q_N$. Under the reference,

$$X_N \mid X_0 = x \sim \mathcal{N}(\Phi x, Q).$$

For zero-mean Gaussian endpoint marginals $X_0 \sim \mathcal{N}(0, V_0)$ and $X_N \sim \mathcal{N}(0, V_N)$, the entropy-optimal endpoint cross-covariance is

$$C_{0N} = \frac{2\Phi V_0 V_N}{Q + \sqrt{Q^2 + 4\Phi^2 V_0 V_N}}. \tag{4.4}$$

This expression is numerically stable even when Φ is small. It reduces to the Brownian scalar coupling in the random-walk limit $a_0 \to 1$ with $r_0 = q/N$.

4.3 Exact covariance grid

Let $b_k = a_0^{N-k}$ and define

$$G_k = \frac{b_k Q_k}{Q}, \qquad \alpha_k = \phi_k - G_k \Phi, \qquad \beta_k = G_k. \tag{4.5}$$

Conditioned on both endpoints, the reference bridge admits the representation

$$X_k = \alpha_k X_0 + \beta_k X_N + \zeta_k, \tag{4.6}$$

where ζ_k is independent of the endpoint pair and has conditional variance

$$S_k = Q_k - \frac{(b_k Q_k)^2}{Q}. \tag{4.7}$$

Consequently,

$$\boxed{V_k = \alpha_k^2 V_0 + \beta_k^2 V_N + 2\alpha_k \beta_k C_{0N} + S_k.} \tag{4.8}$$

Unlike the Brownian quadratic, Equation (4.8) generally has an OU-shaped dependence on physical step. It automatically accounts for persistence or contraction in the identified plant.

4.4 Adjacent cross-covariance and the exact bridge transition

The conditional reference-noise cross-covariance between adjacent times is

$$S_{k,k+1} = a_0 Q_k - \frac{(b_k Q_k)(b_{k+1} Q_{k+1})}{Q}. \tag{4.9}$$

The full adjacent cross-covariance is therefore

$$K_{k,k+1} = \alpha_k \alpha_{k+1} V_0 + \beta_k \beta_{k+1} V_N \tag{4.10}$$

$$+ (\alpha_k \beta_{k+1} + \beta_k \alpha_{k+1}) C_{0N} + S_{k,k+1}. \tag{4.11}$$

Every pair (X_k, X_{k+1}) is jointly Gaussian. The exact Markov realization of the bridge is

$$\boxed{X_{k+1} = a_k^\star X_k + \epsilon_k^\star, \quad a_k^\star = \frac{K_{k,k+1}}{V_k}, \quad r_k^\star = V_{k+1} - (a_k^\star)^2 V_k.} \tag{4.12}$$

The Schur complement guarantees $r_k^\star \geq 0$. Thus no separate one-step projector is required to invent a coupling: the finite-horizon bridge already contains it.

Reference consistency

Equations Equations (4.8) and (4.12) are the reference-consistency change introduced in v0.2 and retained in v0.3. The path, temporal memory, and innovation all come from the same reference. Receding-horizon replanning is simply the repeated solution of the same finite-horizon problem from the newly observed state.

4.5 Stable reference policies and hysteresis

A reference process is an experimental choice, not "whatever the controller happened to do last." If the realized command is fed directly back as the next KL reference, early actuator saturation can become self-reinforcing: the next projection prefers the branch created by the previous projection. This is command hysteresis and is not a global path-space projection.

The preferred policies are:

1. a fixed declared baseline transition;
2. a callback evaluated at a fixed baseline command under the current plant estimate;
3. a slowly filtered version of that commanded reference.

An observed-current-command policy remains useful as an ablation. Filtering is applied once per control opportunity, not once per candidate horizon, so the feasibility search does not itself alter the reference.

4.6 Remaining-horizon governance

At physical update k, let N_k be the planner's remaining horizon. The planner solves Equations (4.8) and (4.12) from the observed V_k to the declared V_T. If the actuator cannot realize the first transition within tolerance, the planner searches larger horizons. A feasible enlarged horizon asks for a gentler first transition. After applying a command, the chosen horizon is committed as

$$N_{k+1} = \max(N_k^{\text{chosen}} - 1, 0). \tag{4.13}$$

This state prevents an important receding-horizon pathology: repeatedly solving a fresh fixed-length horizon can approach the target asymptotically without ever declaring completion. The physical cost of difficult cooling appears as additional updates and compute.

4.7 Brownian limit and role of the original theory

Set $a_0 = 1$ and $r_0 = q/N$. Then $\phi_k = 1$, $Q_k = kq/N$, and the discrete reference converges to Brownian motion as N grows. Equations Equations (4.8) and (4.12) therefore contain the Brownian bridge as a limit. The earlier peak-skew and phase-law results remain valuable for interpretation, but operational control should use the reference that best approximates the learning or inference channel being actuated.

Chapter 5

Bridge Learning as constrained adaptive covariance steering

5.1 The local controlled plant

For each parameter, activity, residual, or recurrent-state block, write the centered local dynamics as

$$z_{k+1} = a_k(u_k)z_k + \epsilon_k, \qquad \mathbb{E}[\epsilon_k \mid \mathcal{F}_k] = 0, \qquad \mathrm{Var}(\epsilon_k \mid \mathcal{F}_k) = r_k(u_k), \tag{5.1}$$

where u_k collects available controls. The active variance follows

$$V_{k+1} = a_k(u_k)^2 V_k + r_k(u_k). \tag{5.2}$$

In a matrix block,

$$\Sigma_{k+1} = A_k(u_k)\Sigma_k A_k(u_k)^\top + R_k(u_k). \tag{5.3}$$

The finite-horizon reference bridge of Chapter 4 supplies a desired first transition $(a_k^\star, r_k^\star)$ or $(A_k^\star, R_k^\star)$. The adapter does not reinterpret bridge marginal width as learning rate. It solves an actuator-realization problem.

5.2 Active width, centering, and robust scale

The centered state $z_k = x_k - \bar{x}_k$ is observer-relative. In practice, $\bar{x}_k$ may be an exponential moving center

$$\bar{x}_{k+1} = (1 - \alpha_c)\bar{x}_k + \alpha_c x_{k+1}. \tag{5.4}$$

The choice of α_c changes the measured width, contraction, and innovation. It is therefore part of the plant model. A slow center reclassifies mean motion as fluctuation; a fast center may absorb genuine low-frequency stochastic state.

For a block metric $M_k \succeq 0$, a conventional active width is

$$V_k = \frac{1}{d}\,\mathrm{tr}(M_k \Sigma_k). \tag{5.5}$$

Useful metrics include optimizer-preconditioned Euclidean coordinates, Fisher or Gauss–Newton traces, precision-weighted predictive-coding errors, and residual-space metrics.

Ordinary variance can be jump-dominated under heavy-tailed gradient or recurrent noise, a concern supported by empirical studies of stochastic-gradient tails in deep networks [21]. The software therefore treats the scale estimator as pluggable. Examples include

$$V_{\text{trim}} = \text{variance after symmetric tail trimming,} \tag{5.6}$$

$$V_{\text{IQR}} = \left(\frac{\text{IQR}}{1.349}\right)^2, \tag{5.7}$$

$$V_{\text{MAD}} = \left(1.4826\,\text{median}|z - \text{median}(z)|\right)^2. \tag{5.8}$$

The Gaussian bridge interpretation is then approximate, but the controlled scale remains robust. Shadow runs should log robust and ordinary widths together; a large unstable ratio is evidence against a Gaussian second-moment model.

Timescale convention

The center update frequency, transition-observer window, and control interval must be reported. A conservative deployment updates the center every plant step, estimates transitions over several center mixing times, and changes controls more slowly than those observer dynamics. Otherwise the controller may heat the plant, induce center lag, observe an artificial width increase, and cool in a self-generated oscillation.

5.3 From exact desired transition to feasible controls

Let $\mathcal{U}_k$ be the admissible control set. The generic realization problem is

$$\min_{u\in\mathcal{U}_k} d_A\big(A_k(u), A_k^\star\big)^2 + \lambda_R d_R\big(R_k(u), R_k^\star\big)^2 + \lambda_u\|u - u_{k-1}\|^2 + \lambda_c\,\text{cost}(u). \tag{5.9}$$

The first two terms match the bridge transition; the third suppresses command chatter; the fourth prices physical compute or other shared resources. A command is feasible only if the predicted next covariance lies within declared absolute and relative tolerances.

5.3.1 Backpropagation or local weight learning

For a locally quadratic scalar block,

$$z_{k+1} = (1 - \eta_k h_k) z_k + \eta_k \xi_k, \qquad \text{Var}(\xi_k) = \frac{v_k}{m_k}, \tag{5.10}$$

so

$$a_k(\eta_k) = 1 - \eta_k h_k, \qquad r_k(\eta_k, m_k) = \frac{\eta_k^2 v_k}{m_k}. \tag{5.11}$$

Given $a_k^\star$, the nominal step is

$$\eta_k^{\text{des}} = \frac{1 - a_k^\star}{\widehat{h}_k}. \tag{5.12}$$

Given a realized step, the nominal effective batch is

$$m_k^{\text{des}} = \frac{\eta_k^2 \widehat{v}_k}{r_k^\star}. \tag{5.13}$$

The division of labor is structural: step size changes both contraction and noise injection, while batch or accumulation changes innovation without directly changing deterministic contraction.

5.3.2 Predictive coding and fixed-point relaxation

A linearized error or residual channel may have

$$e_{j+1} = (1 - \alpha_j h_j)e_j + \sigma_j \xi_j. \tag{5.14}$$

The relaxation step or damping realizes contraction; explicit state noise realizes innovation. If explicit noise is scientifically inappropriate, a step-only actuator can still project onto the reachable set, but it loses the clean separation.

5.3.3 Itinerant and recurrent dynamics

For a discretized recurrent channel, integration step, damping, inhibition, fatigue, or leak may affect contraction, while explicit process noise affects innovation. In a stable heteroclinic channel, innovation also changes saddle dwell time, so the controlled observable cannot be only transversal variance. Dwell-time and itinerary statistics must be reported alongside the bridge scale.

5.4 Curvature endogeneity and edge-of-stability control

The frozen-curvature inversion Equation (5.12) is unreliable when sharpness responds to the command, especially in edge-of-stability regimes [19]. A local first-order model is

$$h(\eta) = h_0 + s_h(\eta - \eta_0). \tag{5.15}$$

Solving $1 - \eta h(\eta) = a^\star$ gives

$$s_h \eta^2 + (h_0 - s_h \eta_0)\eta - (1 - a^\star) = 0. \tag{5.16}$$

The feasible nonnegative root closest to the previous command is selected. This is a local response model, not a complete theory of sharpness adaptation.

Independently, impose a geometry or trust-region cap

$$\eta_k \le \min\left\{\eta_{\max}, \frac{c}{\widehat{h}_k}, \eta_k^{\text{trust}}\right\}, \qquad c < 2. \tag{5.17}$$

If $\eta_k \widehat{h}_k$ approaches the configured edge-of-stability threshold, the controller should slow or freeze bridge progress and recalibrate the response model. It should not interpret a curvature estimate that chases its own command as exogenous truth.

5.5 Reachability, resource quantization, and compute price

At fixed h, ν, and m, the requested next variance W satisfies

$$W = (1 - \eta h)^2 V + \eta^2 \frac{\nu}{m}. \tag{5.18}$$

This convex parabola has minimum

$$W_{\min} = \frac{(\nu/m)V}{h^2 V + \nu/m}. \tag{5.19}$$

Targets below $W_{\min}$ are not reachable in one step. With fixed η, batch bounds imply

$$(1 - \eta h)^2 V + \frac{\eta^2 \nu}{m_{\max}} \le W \le (1 - \eta h)^2 V + \frac{\eta^2 \nu}{m_{\min}}. \tag{5.20}$$

The horizon governor enlarges the remaining bridge horizon until its first transition lies in this set or until a declared maximum multiplier is reached.

Effective batch is not a free continuous actuator. Let $\mathcal{M}$ be the discrete feasible set of accumulation or batch choices. A practical coordinator solves

$$m_k = \arg\min_{m \in \mathcal{M}} \sum_b w_b \left(\frac{\eta_{b,k}^2 v_{b,k}}{m} - r_{b,k}^{\star} \right)^2 + \lambda_c \frac{m}{m_{\text{ref}}} + \lambda_h \left| \log \frac{m}{m_{k-1}} \right| . \tag{5.21}$$

A deadband keeps m_{k-1} when the improvement is too small. This prevents a continuous optimum between two integers from creating a resource limit cycle. Traces report optimizer updates, effective examples or rollouts, solver evaluations, wall clock where available, and intrinsic progress per compute unit.

5.6 Confidence, endogenous estimation error, and bounded probing

A certainty-equivalence analysis that writes

$$e_{k+1} = a_k^2 e_k + \delta_k \tag{5.22}$$

with exogenous δ_k is incomplete. Curvature, noise, width, and contraction are estimated from the process being controlled; their errors are correlated with e_k and the command. Cold cooling regimes are especially difficult because the excitation needed to identify the transition disappears while memory may approach one.

Bridge Learning therefore uses an effective confidence

$$c_k = c_k^{\text{sampling}} c_k^{\text{centering}} c_k^{\text{model}} c_k^{\text{AR}(1)} \in [0, 1]. \tag{5.23}$$

Low confidence blends the proposed command toward a conservative baseline and slows the horizon. In the language of dual control, an input may sometimes be chosen partly to improve identification [20]. Under declared conditions the controller may add a bounded identification probe

$$r_k^{\text{cmd}} = r_k^{\star} + p_k, \qquad 0 \le p_k \le p_{\max}, \qquad \sum_k p_k \le P_{\max}. \tag{5.24}$$

A typical policy activates only when $c_k < c_{\text{probe}}$ and $V_k < V_{\text{cold}}$. Probe innovation is separately accounted and disabled when a validation guard is latched.

A useful local robustness statement is a small-gain condition. Suppose the combined observer/controller disturbance obeys

$$|\delta_k(e_k)| \le L|e_k| + d_k, \qquad |d_k| \le d_{\max}, \tag{5.25}$$

and $|a_k| \le a_{\max}$. If

$$a_{\max}^2 + L < 1, \tag{5.26}$$

then

$$\limsup_k |e_k| \le \frac{d_{\max}}{1 - a_{\max}^2 - L}. \tag{5.27}$$

This is conditional, not a global adaptive-control guarantee. Its value is diagnostic: poor model adequacy or strong control-estimator feedback appears as a large estimated L, at which point active control should be suspended.

5.7 Momentum and augmented-state adequacy

Momentum and Adam introduce second-order memory. The appropriate local object is the companion-form recursion developed in Chapter 6, not a first-order contraction with hidden clipping. Version 0.3 therefore applies the following safety contract:

1. reconstruct the gray-box gradient sample using known step, momentum, and resource commands;
2. compare structural one-step predictions with a forgetting-factor black-box AR(2) fit;
3. monitor contrast uncertainty, regressor conditioning, coefficient slew, Jury margins, damping regime, spectral radius, and Lyapunov conditioning;
4. use antithetic excitation only when the weak contrast is uncertain, the width is cold, and a declared budget remains;
5. refuse companion actuation when structural or curvature adequacy is low.

A full finite-horizon bridge over the augmented (z, v) state remains future work. The current companion actuator is a guarded projection of a scalar bridge transition, so infeasibility may reflect both a genuine momentum floor and mismatch with the desired augmented cross-time covariance.

5.8 Blockwise and shared control

For diagonal blocks,

$$V_{b,k+1} = a_{b,k}^2 V_{b,k} + r_{b,k}. \tag{5.28}$$

Each block can have its own terminal intent, declared reference, remaining horizon, step control, confidence, and probe budget. Shared controls such as effective batch or rollout size are coordinated by Equation (5.21). This is essential in modular systems: a solved routing or edge head should not be forced to share the same contraction and terminal temperature as a fragile execution subsystem.

A scalar global learning rate traces a one-dimensional curve through the high-dimensional space of possible covariance transitions. Generic matrix-valued targets are unreachable without blockwise step multipliers, preconditioning, or richer actuator structure.

5.9 Covariance is a controlled statistic, not a sufficient statistic

Two processes can share the same marginal variance path and differ in:

- cross-time correlation and memory;
- anisotropy of innovation covariance;
- escape and return rates between basins;
- sharpness and local Hessian spectrum;
- the gap between a stochastic-weight average and the current iterate;
- policy divergence, representation change, or itinerary structure.

The bridge transition controls more than a marginal path, but a block-scalar Gaussian transition still does not capture all mechanisms associated with large learning rates or generalization. Therefore tracking RMSE is a controller metric, not a task-performance metric. Experiments must separately report task utility, compute, sharpness or SWA diagnostics, basin or trajectory statistics, and calibration.

Chapter 6

Dynamic sharpness, momentum, and persistent excitation

This chapter develops the v0.3 extension. The covariance bridge remains the slow reference trajectory, while sharpness and momentum define a low-dimensional local plant against which candidate transitions are tested. Planning remains a sequence of scalar recursions, interval propagation, discrete resource search, and small companion-matrix diagnostics rather than a joint nonlinear bridge over all plant states.

6.1 Why a static response curve has the wrong ontology

A local model

$$h(\eta) = h_0 + s_h(\eta - \eta_0)$$

can patch mild curvature of an instantaneous command-response graph inside a very small trust region. It cannot represent three distinct phenomena:

1. *response curvature*: the instantaneous graph is not exactly linear;
2. *dynamics*: sharpness responds with lag and depends on accumulated training progress;
3. *an edge attractor*: realized sharpness can hover near a stability threshold that itself moves with η and momentum μ.

The last two failures are structural. A higher-order polynomial in the current command still treats a dynamical state as an instantaneous constitutive curve. Edge-of-stability observations, self-stabilization analyses, progressive-sharpening models, and central-flow descriptions instead motivate a slow state constrained by the optimizer's stability boundary [19, 22, 23, 24, 25].

6.2 Heavy-ball companion dynamics and the stability triangle

For a locally quadratic direction with heavy-ball momentum,

$$z_{k+1} = (1 + \mu_k - \eta_k h_k)z_k - \mu_k z_{k-1} + \eta_k \xi_k. \tag{6.1}$$

Define

$$c_{1,k} = 1 + \mu_k - \eta_k h_k, \qquad c_{2,k} = -\mu_k, \qquad A_k = \begin{bmatrix} c_{1,k} & c_{2,k} \\ 1 & 0 \end{bmatrix}. \tag{6.2}$$

The characteristic polynomial is $s^2 - c_1 s - c_2$. Jury conditions for both roots to lie in the open unit disk are

$$1 - c_1 - c_2 > 0, \qquad 1 + c_1 - c_2 > 0, \qquad 1 + c_2 > 0. \tag{6.3}$$

Substitution gives the momentum edge

$$0 < \eta h < 2(1 + \mu), \qquad 0 \le \mu < 1. \tag{6.4}$$

The controller works inside a margin-shrunk triangle and uses

$$m_k = 2(1 + \mu_k) - \eta_k h_k \tag{6.5}$$

as the natural edge-distance variable.

The damping boundary is

$$c_1^2 + 4c_2 = 0. \tag{6.6}$$

In the underdamped region, the roots are complex conjugates with product μ, hence

$$\rho(A) = \sqrt{\mu}. \tag{6.7}$$

Learning rate changes the oscillation angle but not the asymptotic pole radius while the block remains underdamped. Momentum is therefore a reachability actuator rather than only a smoothing preference.

6.3 The sharpness plant

The minimal phenomenological model used by the library is

$$h_{k+1} = h_k + \gamma_p g_k - \gamma_r (\eta_k h_k - \theta_k)_+ + e_k, \qquad \theta_k = 2(1 + \mu_k) - \delta. \tag{6.8}$$

Here $g_k \ge 0$ is a declared progress signal such as squared preconditioned update norm, γ_p is the progressive-sharpening rate, γ_r is a fast restoring rate after the margin is pierced, δ is a safety offset, and e_k absorbs process and model error.

Given consecutive sharpness estimates, identification is linear in the two rates. Set

$$y_k = \widehat{h}_{k+1} - \widehat{h}_k, \qquad \varphi_k = \begin{bmatrix} g_k \\ -(\eta_k \widehat{h}_k - \theta_k)_+ \end{bmatrix}, \qquad \beta = \begin{bmatrix} \gamma_p \\ \gamma_r \end{bmatrix}. \tag{6.9}$$

Then $y_k \approx \varphi_k^\top \beta$, so forgetting-factor recursive least squares suffices. The state and parameters can be updated online without embedding sharpness inside the bridge state.

6.4 Robust curvature funnels

A point forecast is not a safety certificate. Version 0.3 propagates

$$h_k^- \le h_k \le h_k^+ \tag{6.10}$$

using parameter uncertainty and residual quantiles. Planning is asymmetric:

$$\text{stability is checked at } h_k^+, \qquad \text{promised contraction is evaluated at } h_k^-. \tag{6.11}$$

If the funnel is too wide to satisfy both statements, the planner extends the horizon, holds the block, falls back to trust-region-only movement, or spends a bounded identification probe. This turns epistemic uncertainty into an explicit control state.

6.5 Two-timescale planning

Sharpness is slow relative to the bridge transition. `SharpnessAwareReferenceBridgePlanner` therefore treats it as a predicted disturbance rather than a jointly steered state. For each candidate remaining horizon it:

1. solves the reference-consistent covariance bridge;
2. realizes each candidate contraction using the current lower sharpness bound;
3. checks edge, Jury, spectral-radius, and conditioning limits at the upper bound;
4. rolls the scalar sharpness recursion forward under the candidate commands;
5. propagates variance and lag covariance;
6. rejects a horizon that becomes infeasible later in the rollout.

The apparent fixed point $\eta = (1 + \mu - c_1^\star)/h$ is handled by sequential substitution because future h_k depends on earlier commands, not the current one. No high-dimensional nonlinear solve is introduced.

6.6 Model ladder and discontinuities

No single phenomenological model should be trusted unconditionally. The library maintains three nested candidates:

Static linear response. A local fallback for mild command dependence inside a narrow trust region.

Sharpness plant. The slow progressive-sharpening/restoration model in Equation (6.8).

Gain-scheduled response. Two or three local responses indexed by edge margin and blended smoothly.

The curvature-adequacy observer evaluates normalized one-step error and interval coverage. The model ladder selects the shortest calibrated description. In the software these roles are implemented by `CurvatureAdequacyObserver` and `CurvatureModelLadder`. If none is adequate, the safe floor is trust-region-only movement with bridge progress held or slowed.

Catapult-style loss spikes and abrupt sharpness drops are detected, not fit. `SharpnessJumpDetector` latches a guard, invalidates the funnel, and forces re-identification.

6.7 Gray-box momentum identification

The commands η_k, μ_k, and resource m_k are known. They should be removed algebraically rather than estimated as constant black-box AR(2) coefficients. Rearranging Equation (6.1) gives

$$\zeta_k = \frac{(1 + \mu_k)z_k - \mu_k z_{k-1} - z_{k+1}}{\eta_k} = h_k z_k + \xi_k, \qquad \mathrm{Var}(\xi_k) = \frac{\nu_k}{m_k}. \tag{6.12}$$

This is the reconstructed local gradient sample. Weighted regression of ζ_k on z_k estimates curvature, and residuals weighted by m_k estimate unit-resource noise. Closed-loop commands are predetermined with respect to the next martingale innovation, so feedback alone does not create ordinary least-squares bias. Noisy centered-state observations do create errors-in-variables; deeper lags can be used as instruments when their validity assumptions hold.

A forgetting-factor black-box AR(2) fit remains useful as an adequacy diagnostic. The software's structural-adequacy observer compares gray-box and black-box one-step predictions. Disagreement flags fast preconditioner motion, non-quadratic drift, or structural misspecification. The implementation class is `StructuralAR2AdequacyObserver`.

6.8 Persistent excitation of order two

The AR(2) estimator regresses z_{k+1} on (z_k, z_{k-1}). In a cold, highly correlated regime the weak eigendirection of the regressor moment matrix is approximately $(1, -1)$, corresponding to the contrast $c_1 - c_2$. Increasing white-probe amplitude scales both state variance and innovation and does not repair the temporal geometry in the idealized stationary limit.

An alternating pair

$$u_k = +\delta, \qquad u_{k+1} = -\delta \tag{6.13}$$

places spectral mass near the Nyquist frequency and directly excites the starved contrast. It is exactly mean-zero over every committed pair, reducing contamination of the center estimate. `AntitheticProbePolicy` triggers from contrast standard error, regressor condition number, and a cold-width gate; supports two amplitudes; stores an unfinished pair in checkpoints; and keeps external probe innovation separate from ordinary batch/noise innovation.

6.9 Companion stability and non-normality

Schur stability alone is not a finite-time covariance guarantee. The augmented covariance error obeys

$$E_{k+1} = A_k E_k A_k^\top + \Delta_k. \tag{6.14}$$

A companion matrix can be strongly non-normal near repeated roots, so Euclidean error may grow transiently even when $\rho(A) < 1$.

Choose $\bar{\rho} \in (\rho(A), 1)$ and solve

$$\bar{\rho}^2 P - A^\top P A = Q, \qquad P \succ 0, \tag{6.15}$$

for a declared $Q \succ 0$. In the norm

$$\|E\|_P = \left\| P^{1/2} E P^{1/2} \right\|_F, \tag{6.16}$$

nominal propagation contracts by at most $\bar{\rho}^2$. If

$$\|\Delta_k\|_P \le L_P \|E_k\|_P + d_{\max}$$

and

$$\bar{\rho}^2 + L_P < 1, \tag{6.17}$$

then a conditional Euclidean bound carries the non-normality penalty

$$\sqrt{\kappa(P)}\, \frac{d_{\max}}{1 - \bar{\rho}^2 - L_P}, \qquad \kappa(P) = \frac{\lambda_{\max}(P)}{\lambda_{\min}(P)}. \tag{6.18}$$

Version 0.3 exposes spectral radius, Jury margins, damping regime, $\kappa(P)$, common-Lyapunov checks over finite trust-region grids, and a matrix-slew recommendation.

6.10 Momentum reachability

In the underdamped region, Equation (6.7) implies an asymptotic covariance-contraction floor of approximately μ per update in the dominant oscillatory mode. Reducing variance by $r = V_T / V_0$ requires at least

$$N \ge \left\lceil \frac{\log r}{\log \mu} \right\rceil, \qquad 0 < r < 1, \tag{6.19}$$

while momentum remains fixed and the approximation holds. Process noise, non-normality, command slew, and cross-time-coupling constraints can only increase the required horizon.

`MomentumStepBatchActuator` searches a bounded momentum grid. For each candidate it derives a step from the requested contraction, checks the curvature funnel asymmetrically, recomputes companion roots and Lyapunov conditioning, and selects one shared effective batch from a discrete compute-priced set. After any shared scaling, all stability diagnostics are recomputed: a smaller step can move a pole toward $+1$, so "smaller learning rate" is not synonymous with "more contractive" in a momentum system.

6.11 Shared dominant sharpness

Per-block sharpness estimates are not independent because the leading Hessian direction belongs to the joint loss. A conservative shared channel uses

$$U = \max_b \frac{\eta_b h_b^+}{2(1+\mu_b)-\delta} \tag{6.20}$$

or a soft maximum. `DominantSharpnessCoordinator` scales participating commands if shared utilization would exceed a declared limit and then forces every block to recompute companion diagnostics. This is a safety guard, not a full coupled Hessian-state model.

Chapter 7

A practical general-purpose algorithm

7.1 User-facing intent

The framework should not expose a renamed warm-up fraction and peak learning rate. A practical channel specification contains:

1. a block or state definition;
2. an active-width observable and centering convention;
3. a declared baseline reference command or transition;
4. a terminal intent or terminal width;
5. a nominal physical horizon and maximum extension;
6. available movement, stochastic, and compute actuators;
7. trust-region, saturation, probe, and validation limits.

For user-facing configurations, terminal intent can be expressed semantically:

Converge. Seek the lowest calibrated and physically reachable terminal width.

Average. Retain a nonzero terminal width suitable for sampling or stochastic-weight averaging.

Continue. Preserve an operating width for continued training, adaptation, or policy exploration.

The software cannot infer task utility from these words alone; an endpoint policy must translate them into a measurable width and validation tolerance.

7.2 Calibration phase

Before active control, run a conservative baseline and collect enough history to answer five questions.

7.2.1 Is the controlled scale meaningful?

Choose ordinary variance and at least one robust scale. Record the center filter and compare its mixing time with the observer and control intervals. Reject blocks whose active width is dominated by center lag, rare jumps, or insufficient sample size.

7.2.2 What local memory order is adequate?

Estimate one-step contraction and innovation under the baseline command. For zero-momentum SGD, a block-scalar AR(1) model may be adequate. For heavy-ball momentum or Adam-like first moments, reconstruct the gray-box gradient sample and compare its companion-form prediction with a black-box AR(2) fit. Active companion control requires declared structural-adequacy, conditioning, and coefficient-slew thresholds. If those tests fail, hold the baseline, fall back to trust-region-only movement, or use a richer augmented-state implementation.

7.2.3 Does a dynamic sharpness model calibrate?

Use naturally occurring warm-up variation and, when necessary, bounded probes to fit Equation (6.8). Compare it with the static and gain-scheduled alternatives using one-step prediction error and interval coverage. Establish an independent trust-region cap. A wide curvature funnel, poor coverage, a detected jump, or operation too close to the momentum edge forces shadow or trust-region-only control rather than certainty-equivalent inversion.

7.2.4 What is the reference process?

Select a conservative baseline command u_{ref} and evaluate

$$(a_0, r_0) = (a(u_{\text{ref}}), r(u_{\text{ref}})) . \tag{7.1}$$

A fixed reference is simplest for a controlled experiment. A slowly filtered callback reference can follow genuine nonstationarity while avoiding last-command hysteresis.

7.2.5 What is the terminal width?

A reachable stationary floor under minimum step and maximum resource is one operational construction. For the scalar model,

$$V_{\text{floor}} = \frac{\eta_{\min}(\nu/m_{\max})}{h(2-\eta_{\min}h)} \tag{7.2}$$

when $0 < \eta_{\min}h < 2$. A task-oriented alternative uses perturbation tolerance:

$$V_T = \sup\left\{v : \mathbb{E}_{\delta_v}[L_{\text{val}}(\theta+\delta_v)] - L_{\text{val}}(\theta) \le \delta_{\text{term}}\right\} . \tag{7.3}$$

The terminal construction, not only its numerical value, must be logged.

7.3 Operational budget accounting

The Brownian parameter q is useful in the explanatory theory but should not be presented as an automatically reduced tuning problem. In the reference-consistent controller, the declared quantities are (a_0, r_0), terminal width, and remaining physical horizon. The operational stochastic expenditure is

$$B_k^{\text{innovation}} = \sum_{j<k}\sum_b r_{b,j}^{\text{realized}}, \qquad B_k^{\text{probe}} = \sum_{j<k}\sum_b p_{b,j}. \tag{7.4}$$

Physical compute is reported separately, for example

$$C_k = \sum_{j<k}\left(\frac{m_j}{m_{\text{ref}}} + \lambda_{\text{solver}} n_j^{\text{sweeps}} + \lambda_{\text{rollout}} n_j^{\text{rollout}}\right). \tag{7.5}$$

Replanning does not conserve a mysterious scalar diffusion budget; it resolves the remaining endpoint problem under the current reference, horizon, probe allowance, and compute limits.

7.4 Reference-consistent channel step

At a control opportunity, perform the following steps.

Algorithm 1: Reference-consistent Bridge Learning channel step

Input: channel state, current observation, terminal width, reference policy, planner, actuator, guards

Output: control command and updated controller state

Update centered or robust width, local dynamics, uncertainty, and model-adequacy diagnostics;
Evaluate the declared or filtered reference transition (a_0, r_0);
Set the nominal remaining horizon N_k;
for *candidate horizons* $N \geq N_k$ *in increasing order* **do**
    solve the exact linear-Gaussian bridge from current width to terminal width;
    take its first exact transition $(a^\star, r^\star)$;
    add a bounded probe only if the confidence policy requests one;
    project $(a^\star, r^\star)$ onto feasible controls, including trust and compute costs;
    if *the command is feasible* **then**
        accept the command and chosen horizon; break
if *no exact command is feasible* **then**
    hold intrinsic progress or use the lowest-error approximation only if explicitly permitted;
Apply validation and safety guards;
Emit predicted next width, confidence interval, controls, saturation, compute, and probe diagnostics;
After the plant step, observe the realized width, update calibration, and commit $N_{k+1} = N - 1$;

The planner's progress is a summary of the committed remaining horizon. It is not assumed to be linear in optimizer update count. A difficult cooling phase may consume many more physical updates than the nominal horizon.

7.5 Shared-resource coordination

With blockwise step sizes but one global effective batch, each block requests an innovation $r_b^\star$. The compute-priced quantized solve in Equation (5.21) produces one shared resource command. For reinforcement learning, the analogous resource may be rollout count or critic minibatches. For fixed-point inference, it may be solver sweeps. For CAROM itinerant dynamics, it may include GLV integration steps.

A coordinator should expose both the continuous optimum and the selected discrete command. This makes quantization error and deadband behavior inspectable. If the resource saturates, the horizon governor should slow progress rather than silently reinterpret the saturated transition as successful.

7.6 Validation and safety guards

A bridge controller can be locally calibrated and still damage the task. A guard monitors a predeclared fixed validation panel and can:

1. hold intrinsic progress;
2. revert movement controls toward the baseline;
3. disable probes;

4. re-estimate the reference and endpoint;
5. latch the channel until manual or automated recovery criteria are met.

For CAROM, the guard should combine endpoint performance with trajectory coverage, itinerary order, and the natural-minus-shuffled causal gap. Edge accuracy alone is insufficient because it remained stable during the observed execution collapse.

For reinforcement learning, guards may use policy KL, constraint violations, critic calibration, and return distributions. For predictive coding, they may use energy descent, residual monotonicity, and reconstruction or task error.

7.7 Failure states and safe fallbacks

The controller should make failure explicit.

Low confidence. Blend toward the baseline command, slow the horizon, and optionally use a bounded probe.

Low structural adequacy. Hold the block or require an augmented-state adapter.

Edge of stability. Apply the trust-region cap and pause bridge acceleration.

Resource saturation. Extend the horizon and report additional compute.

Poor one-step calibration. Return to shadow mode and refit the observer/actuator model.

Validation degradation. Latch the guard and suppress further exploration.

Heavy-tail disagreement. Switch the planning observable to a robust scale while retaining raw variance diagnostically.

7.8 Deployment modes

Three modes are useful.

Shadow. Observe, plan, and log; do not apply commands. This is the mandatory first mode for a new architecture.

Guarded active. Apply commands only to adequate blocks, within conservative bounds and with validation guards.

Research active. Enable probes, adaptive resources, and larger horizon changes under a preregistered experimental protocol.

A checkpoint must include observer filters, centers, reference-policy state, remaining horizons, cumulative innovation and compute, previous commands, guard state, and calibration accumulators.

7.9 Reporting requirements

At minimum, report:

- the observable, center, and robust-scale choice;
- declared reference command and reference-policy dynamics;

- terminal-width construction and nominal/actual horizon;
- one-step calibration and uncertainty coverage;
- structural adequacy and curvature-response diagnostics;
- step, resource, and trust-region saturation;
- realized innovation, probe innovation, examples/rollouts/solver evaluations, and wall clock;
- tracking error and task outcomes;
- sharpness, SWA gap, basin occupation, policy divergence, or trajectory evidence whenever generalization or mechanism is claimed.

Chapter 8

Software architecture of `bridgelearn`

8.1 Design principle: control transitions, not gradients

The library is a control layer around an iterative system, not a replacement optimizer and not a subclass of a learning-rate scheduler. Its generic channel has the form

$$z_{k+1} = A_k z_k + \epsilon_k, \qquad \mathrm{Cov}(\epsilon_k) = R_k. \tag{8.1}$$

A backend-specific adapter observes the local channel and realizes a requested Gaussian transition. The mathematical core does not require a gradient object. Backpropagation, predictive coding, fixed-point relaxation, actor–critic learning, and recurrent dynamics differ primarily in their observers and actuators.

Version 0.3.0 depends only on NumPy in its core. Optional integrations add PyTorch or plotting. The public implementation is block-scalar or diagonal: one active width and one transition per block.

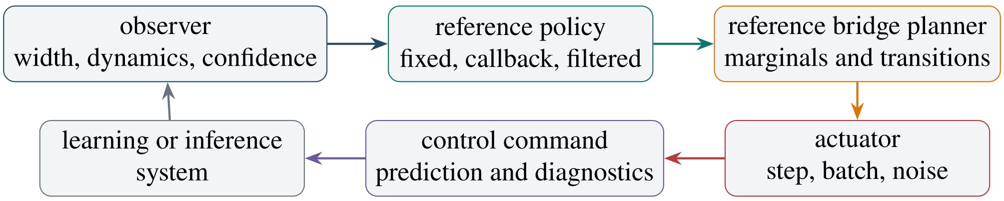


Figure 8.1: BridgeLearn v0.3 control loop. The planner and path use one declared reference process. Realized observations close the loop and populate calibration artifacts.

8.2 Core data model

The principal immutable records are:

`GaussianState` block variances, confidence, and labels;

`GaussianTransition` block contractions and innovation variances;

`LocalDynamics` observed width, current/previous width, lag covariance, curvature center and bounds, unit-resource noise, step, momentum, resource, trust, structural adequacy, coefficient slew, and diagnostics;

`ChannelObservation` state, observed transition, local dynamics, metadata, and application payload;

`ActuationPlan` controls, predicted transition, feasibility, error, and diagnostics;

`ControlCommand` the committed command, target and predicted next widths, progress, and diagnostics;

ChannelState persistent progress, remaining history, cumulative innovation, probe innovation, and compute.

All arrays are validated as finite one-dimensional `float64` values. Scalar inputs broadcast to block count.

8.3 Reference-consistent paths and planners

`LinearGaussianBridgePath` implements Equations (4.4), (4.8) and (4.12). It exposes:

- `covariance_grid()`;
- `adjacent_cross_covariance_grid()`;
- `transition_at(k)`;
- `variance(progress)` and `peak_progress()`.

`ReferenceBridgePlanner` is the recommended receding-horizon object. It selects a reference, solves the bridge from the current observed width to the terminal width, requests its first transition, adds an optional bounded probe, searches longer horizons when actuation is infeasible, and commits the chosen remaining horizon after the plant step.

The reference-policy classes are:

FixedReferencePolicy a fixed declared transition;

CallbackReferencePolicy a transition evaluated from the current observation at a declared baseline command;

FilteredReferencePolicy a slowly filtered contraction and log-innovation reference;

ObservedReferencePolicy the observation's supplied transition, retained for ablation and compatibility.

The legacy `GaussianBridgePath`, `GaussianKLProjector`, and `FeasibilityClock` remain available for Brownian theory and explicit path/reference ablations. They are not the recommended default composition.

8.4 Probing policies

`NoProbePolicy` leaves the exact bridge transition unchanged. `ConfidenceProbePolicy` adds bounded innovation when confidence is below a threshold and the observed width is below a cold threshold. It enforces both per-command and cumulative probe limits. The controller stores probe expenditure separately from ordinary innovation.

A custom probe policy can use richer criteria, but it should preserve the distinction between task-driven exploration and identification energy.

8.5 Actuators

8.5.1 `StepBatchActuator`

This actuator maps target contraction to blockwise step sizes and target innovation to one shared effective batch. It supports:

- step and batch bounds;
- block weights for the shared-resource solve;
- endogenous curvature inversion through `LocalDynamics.step_for_contraction`;
- independent trust-region and stability caps;
- step and batch rate limits;
- integer batch/accumulation quantization;
- compute price and a resource deadband;
- refinement against the one-step Lyapunov equation;
- edge-of-stability, saturation, and compute diagnostics.

8.5.2 `StepNoiseActuator`

This actuator is suited to predictive coding, fixed-point relaxation, and recurrent dynamics with explicit noise. Step size or damping realizes contraction; the square of `noise_std` realizes innovation.

8.5.3 Other actuators

`StepOnlyActuator` is a fixed-resource fallback. `DirectTransitionActuator` exposes contraction and innovation directly to a simulator or specialized dynamical system. Custom actuators implement the same `realize(target, observation)` interface.

8.6 Dynamic sharpness, momentum identification, and stability

Version 0.3 adds a second-order control stack without changing the channel protocol.

`SharpnessPlant` forgetting-factor RLS identification of Equation (6.8);

`CurvatureFunnel` lower/center/upper forecast with confidence and adequacy;

`StaticLinearCurvatureModel` calibrated local fallback;

`GainScheduledSharpnessModel` edge-margin-indexed local response;

`CurvatureAdequacyObserver` one-step error and interval-coverage report;

`CurvatureModelLadder` shortest calibrated model selector;

`SharpnessJumpDetector` latching discontinuity guard;

`DominantSharpnessCoordinator` shared global-edge safety channel;

`GrayBoxMomentumObserver` command-deconfounded curvature and noise identification;

`ForgettingFactorAR2Observer` black-box fallback with an identification-bandwidth slew contract;

`StructuralAR2AdequacyObserver` gray-box/black-box prediction agreement;

`AntitheticProbePolicy` exact committed order-2 probe pairs;

`MomentumStepBatchActuator` joint step, momentum, and discrete resource realization;

`stability.py` Jury geometry, companion roots, damping class, Lyapunov metrics, common-P checks, and conditional small-gain diagnostics.

The planner and actuator use curvature uncertainty asymmetrically. The lower bound maps requested contraction to a command; the upper bound certifies stability. Momentum actuation additionally checks spectral radius, Lyapunov conditioning, coefficient slew, the underdamped floor, and a shared dominant-edge constraint. Probes are not folded into ordinary innovation: the command records total transition, actuator transition, and external signed innovation separately.

8.7 Observers, model adequacy, and robust scales

The library includes:

`SecantCurvatureObserver` positive local secant curvature;

`CurvatureResponseObserver` legacy first-order static response, retained as a local fallback;

`SplitBatchNoiseObserver` block noise from independent gradient halves;

`EmpiricalTransitionObserver` contraction and innovation from paired centered states;

`ResidualDecayObserver` fixed-point or predictive-residual contraction;

`AR2AdequacyObserver` generic AR(1) versus AR(2) residual diagnostic;

`GrayBoxMomentumObserver` structural momentum curvature/noise estimate;

`StructuralAR2AdequacyObserver` structural versus black-box companion adequacy;

`CenteredWidthObserver` explicit EMA center plus pluggable scale estimator;

`VarianceScale`, `TrimmedVarianceScale`, `IQRScale`, `MADScale` ordinary and robust width estimators;

`ExponentialFilter`, `AttackReleaseFilter` symmetric or asymmetric smoothing.

Observers should return confidence rather than silently clip pathological estimates. The controller combines ordinary confidence with curvature and structural adequacy. A low-confidence block stays closer to its baseline, slows its bridge horizon, or becomes trust-region-only.

8.8 Adapters and backend neutrality

`DynamicsAdapter` combines three callbacks:

1. read `LocalDynamics` from an application state;
2. apply controls to that state;
3. realize a target transition with a reusable actuator.

`CallbackAdapter` supports fully custom observation and actuation. Both satisfy the generic channel interface.

The PyTorch integration, `TorchOptimizerAdapter`, associates one bridge block with each optimizer parameter group. The application supplies `TorchGroupStatistics`. For momentum or Adam-like groups, a structural adequacy signal is required by default. The adapter raises if the diagnostic is missing unless the user explicitly opts into an approximate first-order experiment. Adam β_1 control remains disabled unless explicitly enabled. This is a safety gate, not merely a warning.

8.9 Persistent and episodic channels

A `Channel` may be persistent or episodic. Persistent channels survive the whole training run and suit weights, actors, critics, or long-lived recurrent parameters. Episodic channels are deep-copied and reset for each predictive-coding inference problem, fixed-point solve, or recurrent episode.

`BridgeController` owns named channels and their states. Its main methods are:

`plan(name, context)` observe and produce a command;

`apply(name, context, command)` apply controls unless in shadow mode;

`after_step(name, context, command)` commit progress and update accounting;

`done(name)` report terminal progress;

`episode(name)` create an isolated episodic controller;

`state_dict()` and `load_state_dict()` checkpoint state.

8.10 Diagnostics and calibration

`BridgeTrace` records target, predicted, and observed widths, controls, progress, feasibility, saturation, reference diagnostics, probe innovation, and compute. It writes JSON and flat CSV.

`CalibrationReport` computes:

- sample count and block count;
- one-step bias, MAE, and RMSE;
- predicted–realized correlation;
- residual quantiles;
- uncertainty-interval coverage when observed standard errors are supplied.

This artifact is central to the scientific program. A controller whose one-step predictions are uncalibrated cannot support a mechanistic interpretation even when task performance improves.

8.11 Minimal first-order persistent example

For SGD without momentum or for a first-order relaxation plant, the public API remains compact:

```python
from bridgelearn import (
    BridgeController, Channel, DynamicsAdapter,
    FixedReferencePolicy, GaussianTransition,
    LocalDynamics, ReferenceBridgePlanner,
    StepBatchActuator,
)

reference = GaussianTransition(
    contraction=[0.995], innovation=[1e-3]
)
planner = ReferenceBridgePlanner(
    terminal_variance=[5e-3],
    total_steps=200,
    reference_policy=FixedReferencePolicy(reference),
```

```
    max_horizon_multiplier=32.0,
)

def read_dynamics(system):
    return LocalDynamics(
        variance=[system.active_variance],
        curvature=[system.curvature],
        noise_scale=[system.unit_batch_noise],
        step_size=[system.step_size],
        effective_batch=system.effective_batch,
        confidence=[system.observer_confidence],
        trust_region_step=[system.trust_region_step],
    )

adapter = DynamicsAdapter(
    read_dynamics=read_dynamics,
    apply_controls=system.apply_bridge_controls,
    actuator=StepBatchActuator(
        step_bounds=(1e-5, 0.2),
        batch_values=(16, 32, 64, 128, 256, 512),
        compute_price=0.004,
    ),
)
controller = BridgeController({
    "weights": Channel(
        name="weights", adapter=adapter, planner=planner
    )
})
```

Listing 8.1: Reference-consistent first-order channel.

8.12 Momentum shadow configuration

The recommended first deployment for Adam or momentum is shadow mode. The application supplies current and previous width, lag covariance, a curvature funnel, noise, momentum, structural adequacy, coefficient slew, and probe diagnostics.

```
from bridgelearn import (
    AntitheticProbePolicy,
    DominantSharpnessCoordinator,
    FixedReferencePolicy,
    GaussianTransition,
    MomentumStepBatchActuator,
    SharpnessAwareReferenceBridgePlanner,
    SharpnessPlant,
)

sharpness = SharpnessPlant(
    gamma_p=1e-3, gamma_r=0.2,
    safety_offset=0.1, forgetting=0.995,
    uncertainty_scale=2.5,
)
probe = AntitheticProbePolicy(
    contrast_se_threshold=0.04,
    condition_threshold=500.0,
    cold_variance=1e-2,
```

```
    max_realized_variance=2e-5,
    max_total_realized_variance=2e-3,
    amplitude_levels=(0.5, 1.0),
)
planner = SharpnessAwareReferenceBridgePlanner(
    terminal_variance=[5e-3, 5e-3],
    total_steps=4000,
    reference_policy=FixedReferencePolicy(
        GaussianTransition(
            contraction=[0.99, 0.99],
            innovation=[1e-4, 1e-4],
        )
    ),
    probe_policy=probe,
    sharpness_model=sharpness,
    freeze_on_inadequacy=True,
)
actuator = MomentumStepBatchActuator(
    step_bounds=(1e-6, 2e-3),
    momentum_bounds=(0.0, 0.95),
    batch_values=(32, 64, 128, 256, 512),
    control_momentum=True,
    max_spectral_radius=0.995,
    max_lyapunov_condition=1e4,
    dominant_coordinator=DominantSharpnessCoordinator(
        safety_offset=0.1,
        maximum_utilization=0.95,
    ),
)
```

Listing 8.2: Core v0.3 momentum objects.

The controller may compute these commands in `mode="shadow"` without mutating the optimizer. An explicit callback is required to apply a signed probe exactly once.

8.13 Episodic predictive-coding pattern

```
with controller.episode("activities") as inference:
    while not inference.done:
        command = inference.plan(pc_state)
        inference.apply(pc_state, command)

        pc_state.relax_once()

        inference.after_step(pc_state, command)
```

Listing 8.3: Episodic predictive-coding or fixed-point channel.

The episodic planner may use a declared damped-relaxation reference. Its terminal width can act as one stopping criterion, but an absolute residual or energy guard should also be required.

8.14 Package organization

Table 8.1: BridgeLearn v0.3 source organization.

Module	Responsibility
`types.py`	Validated states, transitions, observations, plans, commands, and channel state.
`paths.py`	Brownian legacy paths and exact linear-Gaussian bridge paths.
`references.py`	Fixed, callback, filtered, and observed reference policies.
`planners.py`	Reference-consistent remaining-horizon planner.
`probing.py`	No-probe, bounded confidence, and committed antithetic probe policies.
`actuators.py`	First-order and momentum step/resource/noise realization.
`observers.py`	Generic curvature, noise, transition, residual, adequacy, and filters.
`sharpness.py`	Dynamic sharpness, funnels, adequacy ladder, jumps, and shared edge.
`identification.py`	Gray-box momentum, black-box AR(2), IV, and structural adequacy.
`stability.py`	Companion/Jury geometry, Lyapunov metrics, and small-gain reports.
`scales.py`	Centered ordinary and robust width estimators.
`controller.py`	Persistent and episodic orchestration, shadow mode, accounting, checkpointing.
`diagnostics.py`	Trace export, calibration, and control-effort summaries.
`integrations/torch.py`	Optimizer-group adapter and momentum adequacy gate.

8.15 Tests and extension path

The v0.3 suite contains thirty-eight tests. In addition to unit formulas, it covers delayed and noisy observations, integer-resource control, AR(2) misspecification, heavy-tail contamination, probing, curvature response, exact reference bridges, closed-loop behavior, calibration, checkpointing, and projector invariants.

Current limitations are deliberate: scalar or diagonal widths, no full finite-horizon augmented momentum bridge, and no automatic terminal-intent inference. Future covariance representations can be diagonal, low rank, Kronecker, or full matrix without changing the channel–reference–planner–actuator separation.

Chapter 9

Application patterns

9.1 Backpropagation and language-model training

For ordinary minibatch backpropagation, Bridge Learning wraps an existing optimizer. The base optimizer remains responsible for the mean direction, momentum, preconditioning, clipping, and weight decay. The bridge layer modulates parameter-group learning rates and, where possible, effective batch or accumulation.

A practical block partition for a transformer-like model might include:

- embeddings and input projections;
- attention query/key/value and output projections;
- feed-forward or MLP blocks;
- normalization and gating parameters;
- output or task-specific heads.

The blocks may share a global training context while retaining different terminal widths, declared references, adequacy scores, and remaining horizons. They need not peak or cool at the same physical update, and an inadequate block may remain at its baseline while other blocks advance.

Momentum and Adam moments introduce additional state. Version 0.3 does not silently absorb them into an AR(1) plant. It requires an AR(2)-adequacy diagnostic for active PyTorch use; when adequacy is low, the safe action is to hold the baseline or use an augmented local state,

$$\begin{pmatrix} z_{k+1} \\ v_{k+1} \end{pmatrix} = \mathcal{A}_k \begin{pmatrix} z_k \\ v_k \end{pmatrix} + \epsilon_k, \tag{9.1}$$

where v_k is velocity or optimizer memory. The same covariance-steering architecture then applies to the augmented matrix transition.

9.2 Predictive coding

Predictive-coding systems often alternate or interleave two processes:

1. inference or relaxation of latent activities and prediction errors;
2. local weight updates using the settled or partially settled state.

This naturally maps to two channel scopes.

9.2.1 Episodic activity or error channels

At the start of each input or episode, an activity channel is reset. A target path can move from the feed-forward or initialized error width to a terminal residual width. Desired contraction maps to inference step or damping; innovation maps to explicit state noise, stochastic message updates, or replica averaging. The bridge clock determines how rapidly the episode approaches the terminal neighborhood.

The episode can terminate when both

$$s \geq 1 - \epsilon_s \tag{9.2}$$

and an absolute residual criterion is met. This keeps the bridge from declaring completion merely because its nominal clock ended while the actual fixed-point residual remained large.

Predictive coding is especially compatible with the transition abstraction because local relaxation can often be observed directly without gradients. The empirical transition observer measures contraction and innovation from consecutive activity or error snapshots and lowers confidence when a first-order fit is inadequate.

9.2.2 Persistent synaptic channels

Weights evolve on a slower timescale. Their channels persist across examples and may use local learning rates, minibatch resources, or explicit update noise. The inference channel and weight channel can be coordinated: for example, freeze or slow weight progress when the activity episode fails to settle reliably.

Whittington and Bogacz showed conditions under which predictive-coding-style local updates approximate backpropagation [18]. Bridge Learning does not require that equivalence. It only requires a measurable local transition and an actuator mapping.

9.3 Fixed-point and deep-equilibrium inference

Deep-equilibrium models define representations as fixed points and differentiate implicitly through them [1]. CAROM's fixed-point arm similarly used damped iteration. In the reported screen, residuals decreased monotonically but remained near 0.05 after eight sweeps, indicating contraction without adequate settling [3].

An episodic Bridge Learning channel can control:

- relaxation factor or damping;
- explicit solver noise or regularization;
- solver tolerance;
- maximum sweep count or shared compute budget;
- layerwise or blockwise residual widths.

The target terminal width and declared relaxation reference define a residual-distribution objective rather than a fixed sweep count. Easy inputs may reach the terminal neighborhood quickly; hard inputs receive more sweeps, bounded by a shared compute coordinator. This aligns with the adaptive-computation motivation of fixed-point semantics in the CAROM note [5].

9.4 Reinforcement learning

Backpropagation in reinforcement learning is nonstationary and often resource-limited. Natural persistent channels include:

- policy or actor parameters;
- value function or critic parameters;
- a shared encoder;
- entropy-temperature parameters;
- world-model or dynamics parameters;
- target-network update dynamics.

The stochastic resource m may represent minibatch size, trajectory count, rollout length, number of environments, or number of return samples. The appropriate scaling is algorithm-specific, so the adapter should expose an empirical innovation model rather than assume that all resources reduce variance exactly as $1/m$.

A useful RL pattern is to let the actor and critic have separate covariance paths but coordinate one rollout budget. The critic may require stronger contraction and lower terminal width, while the actor may retain a nonzero terminal temperature to preserve exploration. A validation guard should use return distributions, constraint violations, and policy divergence rather than supervised validation loss alone.

Because data distribution changes with the policy, receding-horizon replanning is essential. Old curvature, noise, and reference estimates should decay or be invalidated after large policy changes. Compute matching must use environment interactions and rollout cost, not optimizer updates alone.

9.5 CAROM

CAROM separates command routing, a reusable operator core, workspace state, and recurrent execution semantics [3, 5]. This modular structure is a strong match for multi-channel control.

9.5.1 Persistent weight blocks

A conservative first partition is:

1. command routing and dependency/edge head;
2. operator attention projections;
3. transform and gating submodules;
4. GLV inhibition, growth, leak, and fatigue parameters;
5. workspace and output readout.

The 12k experiment suggests at least two distinct terminal intents. The edge subsystem was already accurate and should have cooled or frozen; downstream execution remained fragile and needed continued but safer adaptation. A global OneCycle curve could not express that asymmetry [4].

The first active experiment should therefore use blockwise learning-rate channels with a fixed batch, followed by a compute-priced, quantized step-plus-accumulation version. The original OneCycle run and a shadow Bridge Learning run should be retained as controls. Checkpoints should be dense through the old failure window.

9.5.2 Fixed-point workspace channel

For each example or batch, define an episodic residual channel. Use consecutive workspace residuals to estimate contraction, map desired contraction to damping, and end the episode only after the terminal residual width and an absolute tolerance are both satisfied. This turns the number of sweeps into an observable output.

9.5.3 Itinerant control channel

CAROM's itinerant execution uses a GLV competition process over command modes. The reported GPU screen used 70 integration steps, making itinerant variants roughly 17 times more expensive per training step than scheduled execution; the report explicitly identifies learned step count or adaptive stopping as necessary for scaling [3].

Useful episodic observables include:

- transverse activity variance around the intended heteroclinic channel;
- distance from simplex vertices during mode dwell;
- itinerary entropy and route variance across noise resamples;
- dwell-time variance;
- innovation amplitude near saddles;
- terminal halt-mode residual.

The natural realization is `DynamicsAdapter` with `StepNoiseActuator`: integration step or damping controls local contraction; explicit GLV noise controls innovation. The CAROM working note predicts that mean saddle exit time scales approximately as

$$\mathbb{E}[\tau_{\mathrm{exit}}] \propto \frac{\log(1/\eta_a)}{\lambda_u}, \tag{9.3}$$

where η_a is mode-noise amplitude and λ_u is the unstable rate [5]. Noise changes should therefore be rate-limited and evaluated through observed dwell times, not only through covariance tracking.

9.5.4 CAROM validation guard

The guard should combine:

- fixed-corpus endpoint accuracy;
- itinerary order and coverage;
- natural-minus-shuffled endpoint gap;
- transition recall or related causal trajectory metrics;
- per-loss terms and routing entropy.

Edge accuracy alone must not trigger "all clear," because it remained stable during the execution collapse. Guard thresholds, patience, and checkpoint-selection rules should be preregistered across multiple seeds.

9.6 A common channel map

The common structure is not a common optimizer. It is a common transition-control interface.

Table 9.1: Mapping learning paradigms into Bridge Learning channels.

System	Controlled state	Contraction actuator	Innovation or resource actuator
Backpropagation	Preconditioned parameter or update deviation	Group learning rate, damping, weight decay	Batch, accumulation, explicit update noise
Predictive coding	Activities, prediction errors, local synaptic deviations	Inference step, damping, local learning rate	State noise, replica count, minibatch
Deep equilibrium	Fixed-point residual or latent-state deviation	Relaxation, solver damping	Solver noise, compute budget, tolerance
Reinforcement learning	Actor, critic, encoder, target dynamics	Group learning rates, target-update rate	Rollouts, trajectories, environments, minibatch
CAROM fixed point	Workspace residuals	Relaxation and damping	Sweep budget, optional state noise
CAROM itinerant	Mode activities and dwell statistics	Integration step, inhibition/fatigue controls	GLV noise and integration budget

Chapter 10

Reproducible sandbox examples

This chapter reports six small experiments used to develop and test the framework [26]. The first two exercise reference-consistent first-order control. The last four isolate the v0.3 mechanisms for dynamic sharpness, persistent excitation, companion stability, and momentum reachability. All are synthetic mechanism tests. None is evidence of improved language-model, CAROM, predictive-coding, or reinforcement-learning performance.

10.1 Verification environments and artifacts

The reference-consistent first-order examples use the first-order core retained in v0.3. The v0.3 mechanism suite was run with Python 3.13.5, NumPy 2.3.5, Matplotlib 3.10.8, and BridgeLearn 0.3.0. The complete experiment bundle contains the code, a vendored wheel, CSV traces, JSON summaries, plots, checksums, and a clean reproducibility run.

10.2 Reference-consistent stochastic quadratic

The controlled plant was

$$x_{k+1} = (1 - \eta_k h_k)x_k + \eta_k \sqrt{v_k/m_k}\, \xi_k. \tag{10.1}$$

The BridgeLearn arm and comparison OneCycle arm used the same 849 physical updates, mean effective batch 462.70, total example-equivalent count 392,829.6, and matched initial/maximum/final step extrema. A curvature/noise shock tested feedback.

Table 10.1: Compute-matched first-order sandbox result.

Metric	BridgeLearn	Compute-matched OneCycle
Final active variance	0.005268	0.010778
Final mean error	0.04662	0.000899
Final quadratic task loss	0.005023	0.007275
Cumulative task loss	87.044	91.339
Tracking RMSE	4.99×10^{-4}	1.86×10^{-2}

The result is intentionally decomposed: BridgeLearn produced a colder stochastic state, whereas OneCycle produced a more accurate final mean. The combined toy task loss happened to favor BridgeLearn. This is exactly why mean optimization and covariance control must be reported separately.

One-step predictions were calibrated against realized, rather than target, width: bias -1.47×10^{-7}, MAE 2.00×10^{-5}, RMSE 4.49×10^{-5}, and correlation 0.999997.

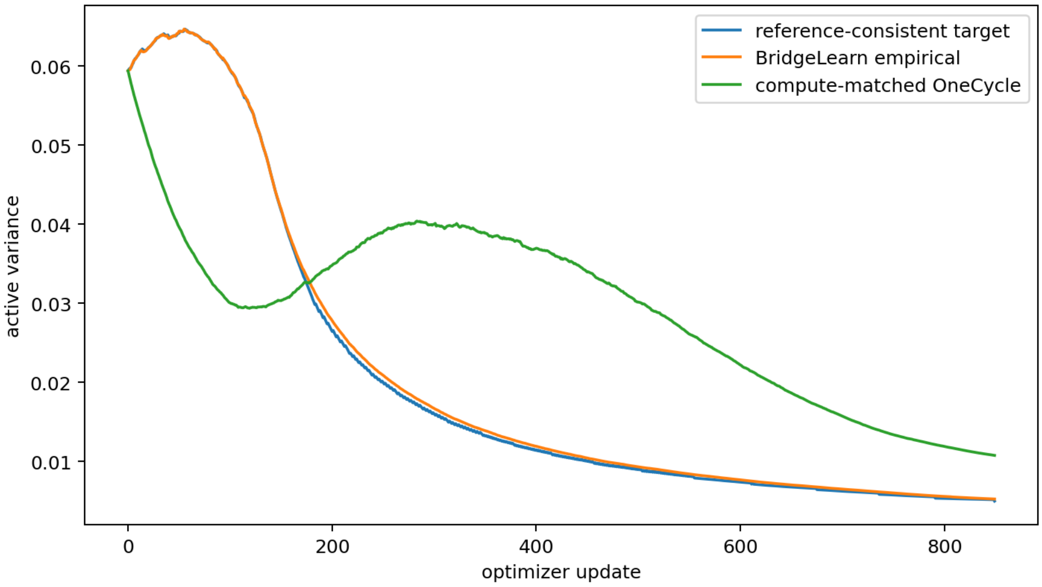


Figure 10.1: Reference-consistent variance tracking in the compute-matched stochastic quadratic.

10.3 Episodic predictive-relaxation channel

An ensemble of prediction errors followed

$$e_{s+1} = (1 - \alpha_s h)e_s + \sigma_s \xi_s. \tag{10.2}$$

An episodic channel selected relaxation step and explicit innovation. Over 60 sweeps, variance-tracking RMSE was 1.26×10^{-5} and final observed variance was approximately 10^{-3}.

10.4 Dynamic sharpness as an identified plant

The synthetic plant followed Equation (6.8) with true $\gamma_p = 0.028$ and $\gamma_r = 0.42$. The command schedule warmed, operated near the edge, cooled, and rewarmed, producing lag and hysteresis.

The identified final values were 0.027041 and 0.415060, with relative errors −3.43% and −1.18%. After a 60-update burn-in,

$$\mathrm{RMSE}_{\text{plant}} = 0.02280, \qquad \mathrm{RMSE}_{\text{static}} = 0.22321, \tag{10.3}$$

so the dynamic plant improved one-step RMSE by 9.79× over a rolling static $h(\eta)$ fit.

At update 150 a counterfactual held $\eta = 0.29$ and $\mu = 0.85$. Current sharpness implied utilization 0.96795, so a frozen-curvature controller would call the command safe indefinitely. The true edge crossing occurred after 15 future updates, the point forecast predicted 17, and the robust upper funnel warned after 4.

10.5 Antithetic order-2 identification

The cold plant was

$$z_{k+1} = 1.8z_k - 0.81z_{k-1} + u_k + \epsilon_k, \tag{10.4}$$

with a repeated pole at 0.9. No probe, known white excitation, and exact $+\delta, -\delta$ pairs were compared over 1,000 trials. White and antithetic probes were calibrated to nearly equal added state width.

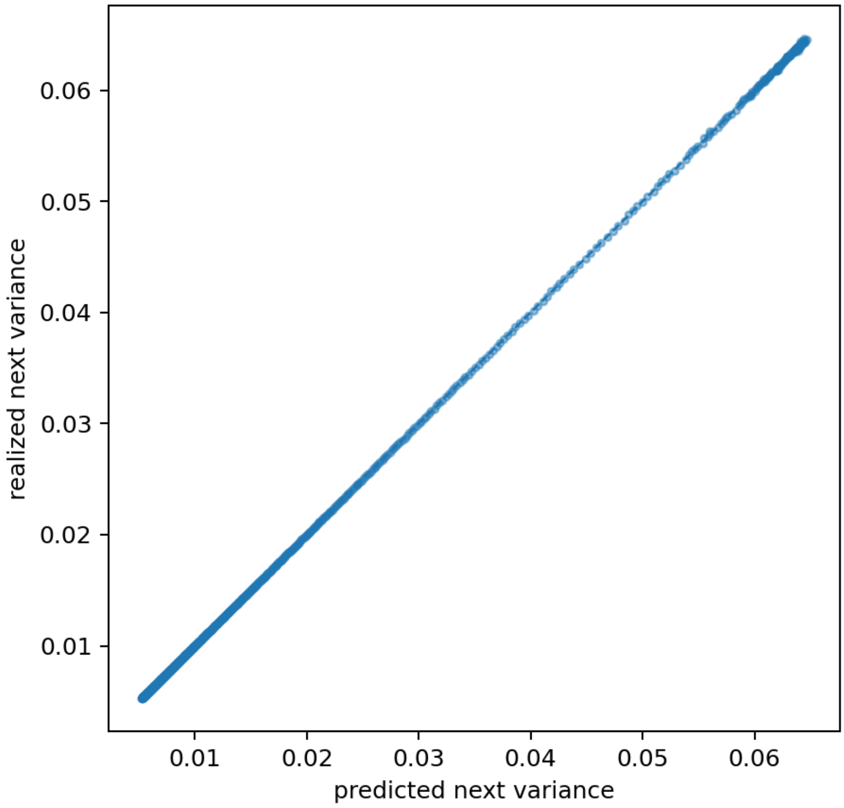


Figure 10.2: Predicted versus realized one-step variance for the first-order example.

Table 10.2: Order-2 identification at matched width disturbance.

Probe	Contrast RMSE	Median cond(M)	Median $\lambda_{\min}(M)$	Added variance
None	0.077219	340.73	2.28×10^{-5}	0
White	0.042051	333.22	7.62×10^{-5}	0.00858
Antithetic	**0.002493**	**2.56**	**0.007810**	0.00999

Relative to white excitation, antithetic excitation reduced contrast RMSE by 16.87×, increased the smallest information eigenvalue by 102.5×, and reduced the condition number by 130.0×.

10.6 Companion stability and non-normality

Two stable matrices had the same spectral radius 0.9:

$$A_{\text{rep}} = \begin{bmatrix} 1.8 & -0.81 \\ 1 & 0 \end{bmatrix}, \qquad A_{\text{under}} = \begin{bmatrix} 0.4 & -0.81 \\ 1 & 0 \end{bmatrix}.$$

Table 10.3: Stable companion systems with the same asymptotic radius.

Case	$\rho(A)$	$\kappa(P)$	State amp.	Covariance amp.	Peak step
Near repeated root	0.9000	5104.96	7.03	**49.47**	9
Underdamped, same radius	0.9000	1.63	1.15	1.32	1

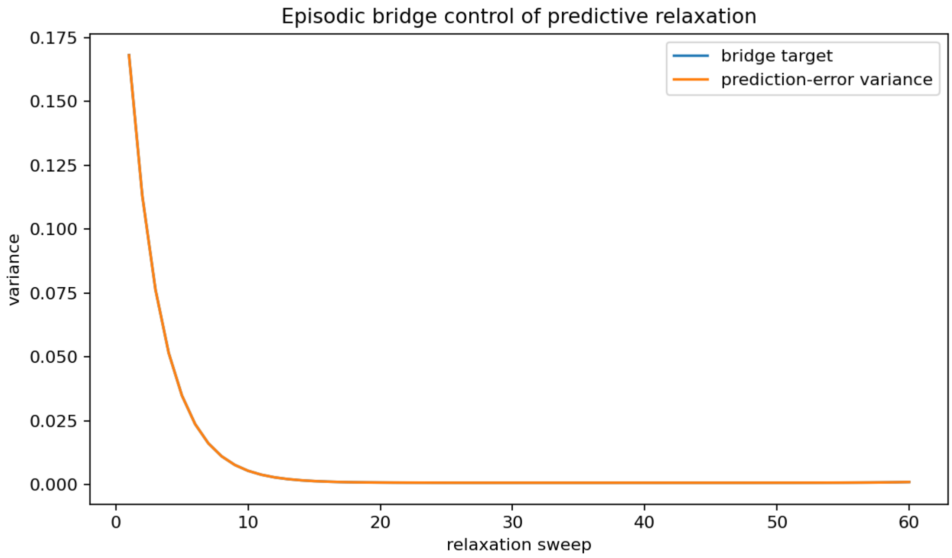


Figure 10.3: Episodic bridge control of a predictive-coding-style relaxation channel.

The near-repeated-root covariance error grew 49.47× in Euclidean norm. At the same update, its Lyapunov-adapted norm had contracted to 0.24280, below the bound 0.24578. Spectral radius alone therefore missed the finite-time risk.

10.7 Momentum-limited reachability

For $V_0 = 0.08$, $V_T = 0.005$, and fixed $\mu = 0.9$, the ideal underdamped floor gives

$$N_{\min} = \left\lceil \frac{\log(0.0625)}{\log(0.9)} \right\rceil = 27 \tag{10.5}$$

updates before noise, non-normality, command slew, or actuator limits are charged.

Five requested one-step variance ratios were infeasible with momentum fixed at 0.9 and all five were feasible when momentum could move. The unconstrained search selected $\mu = 0$; this exposes the geometry but is not a recommended production command because the toy objective omitted retention and slew costs.

10.8 What the sandbox suite establishes

Within the declared local models, the suite supports four narrow conclusions:

1. dynamic sharpness is worth treating as state;
2. probe spectrum matters more than white amplitude in cold momentum regimes;
3. Schur stability is insufficient near non-normal repeated roots;
4. momentum is a physical reachability actuator that can impose a minimum cooling horizon.

It also verifies software composition, checkpointable probe pairs, funnel planning, calibration, discrete resource accounting, and reproducibility.

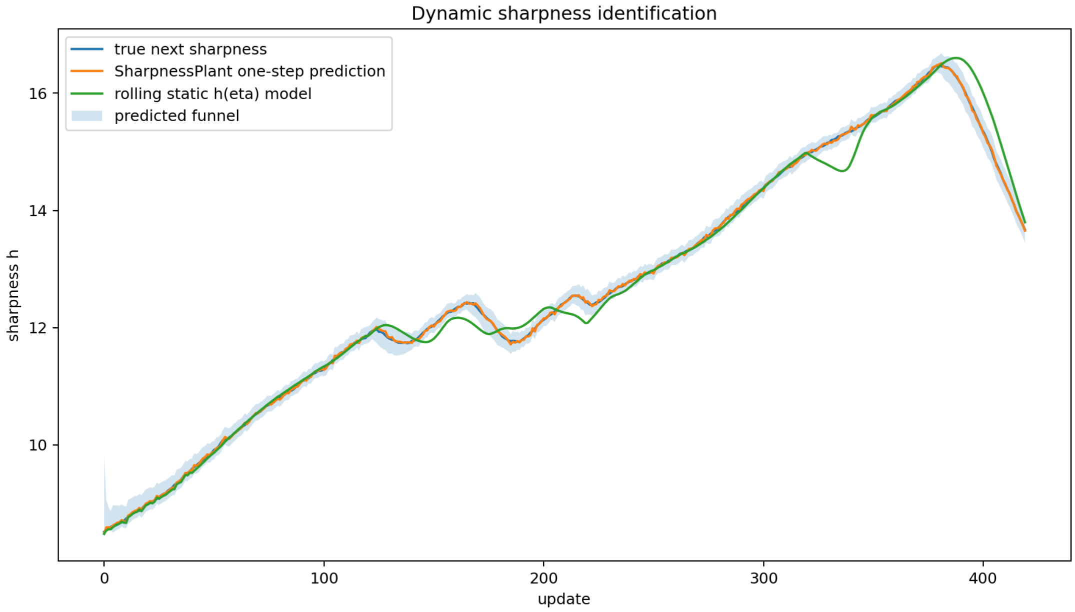


Figure 10.4: Latent sharpness, identified plant forecast, uncertainty funnel, and rolling static response.

10.9 What the sandbox suite does not establish

The examples are scalar or 2×2, synthetic, and partly generated from the same model forms that the library identifies. They do not establish universal sharpness dynamics, valid instruments in every application, improved implicit regularization, better generalization, improved CAROM routing, lower predictive-coding energy, or better RL return. The companion actuator still projects a scalar marginal bridge through an augmented plant rather than solving the exact finite-horizon augmented-state Schrodinger bridge.

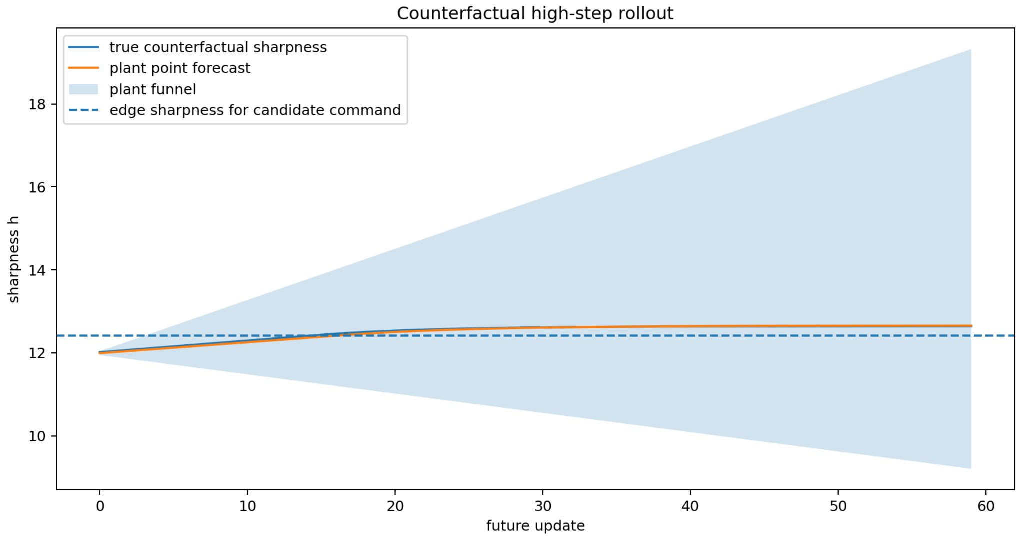


Figure 10.5: Counterfactual edge forecast. The upper funnel provides an intentionally conservative warning.

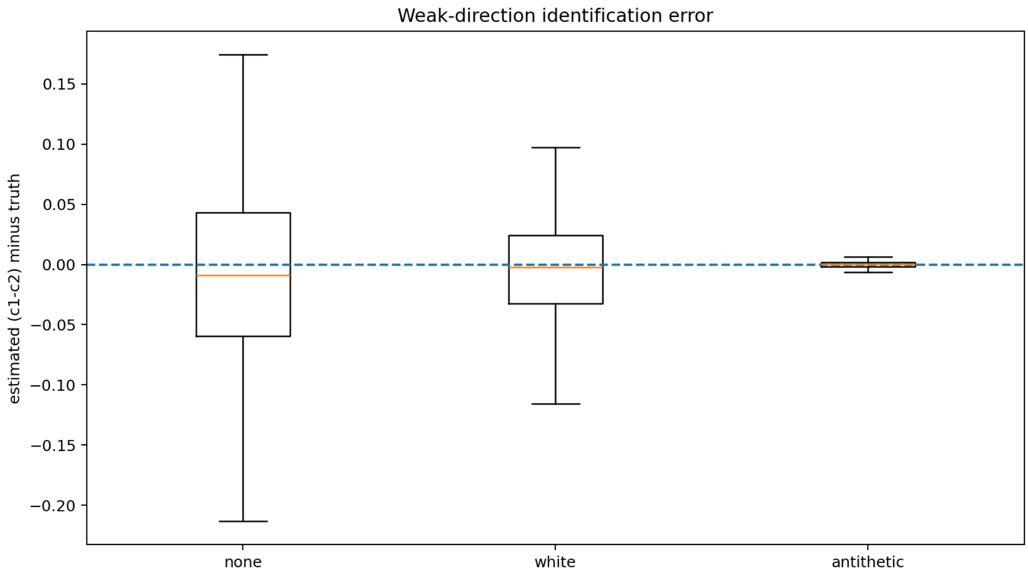
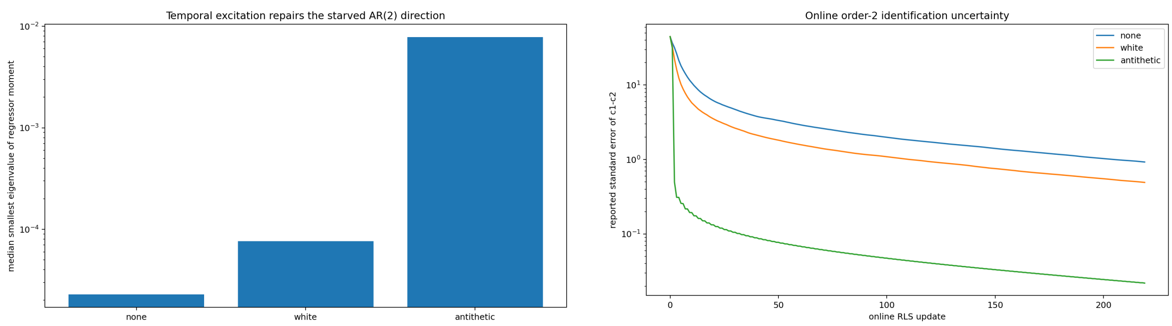


Figure 10.6: Trial distribution of the weak $c_1 - c_2$ contrast error.



(a) Regressor information.

(b) Online contrast standard error.

Figure 10.7: Antithetic excitation targets the starved temporal direction rather than merely adding amplitude.

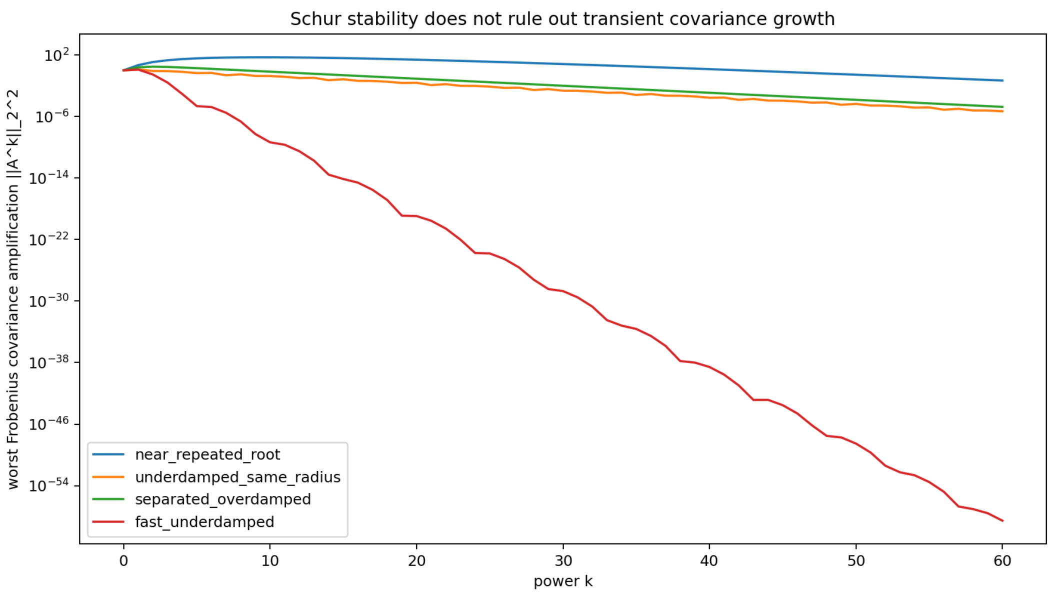


Figure 10.8: Transient Euclidean covariance amplification for stable companion systems.

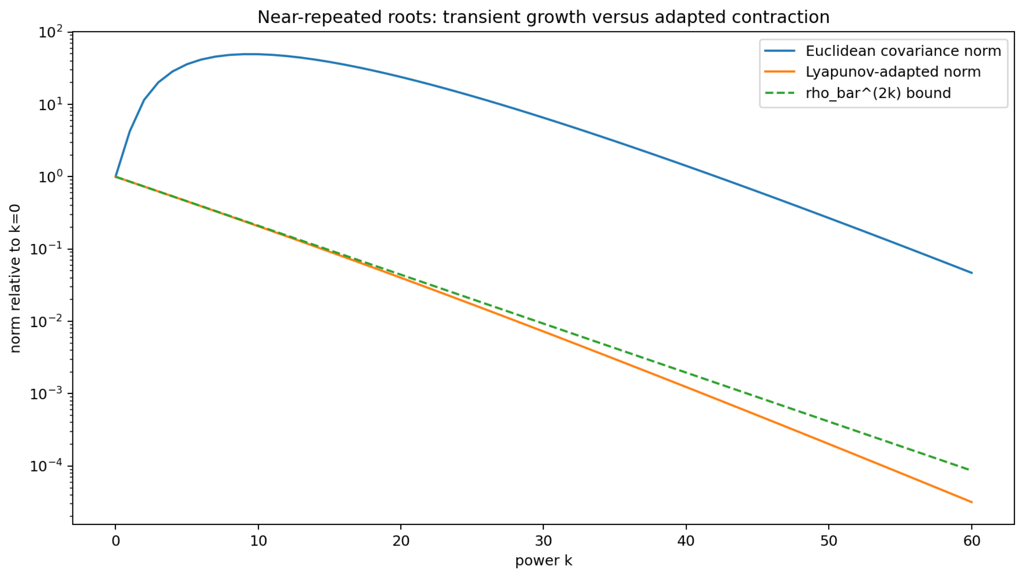


Figure 10.9: Euclidean versus Lyapunov-adapted covariance norms in the near-repeated-root case.

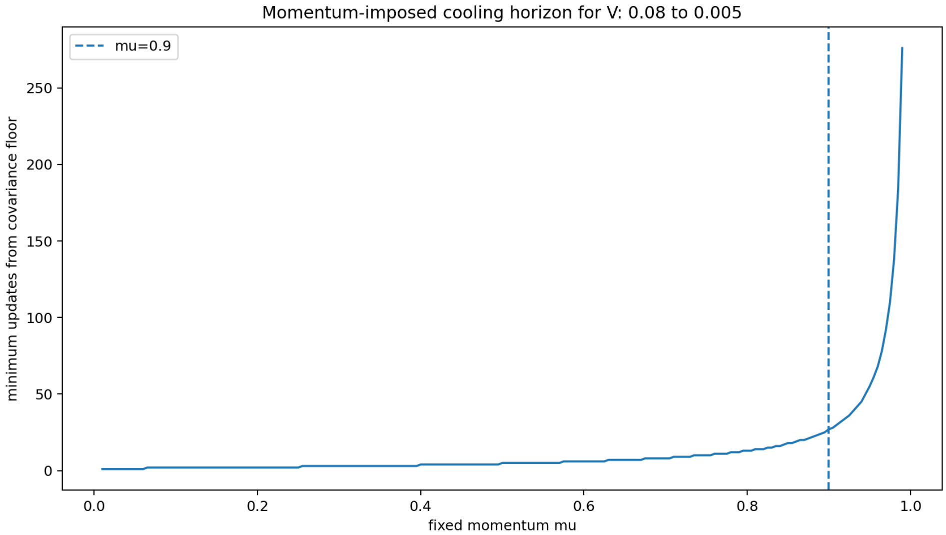


Figure 10.10: Minimum ideal cooling horizon implied by fixed momentum.

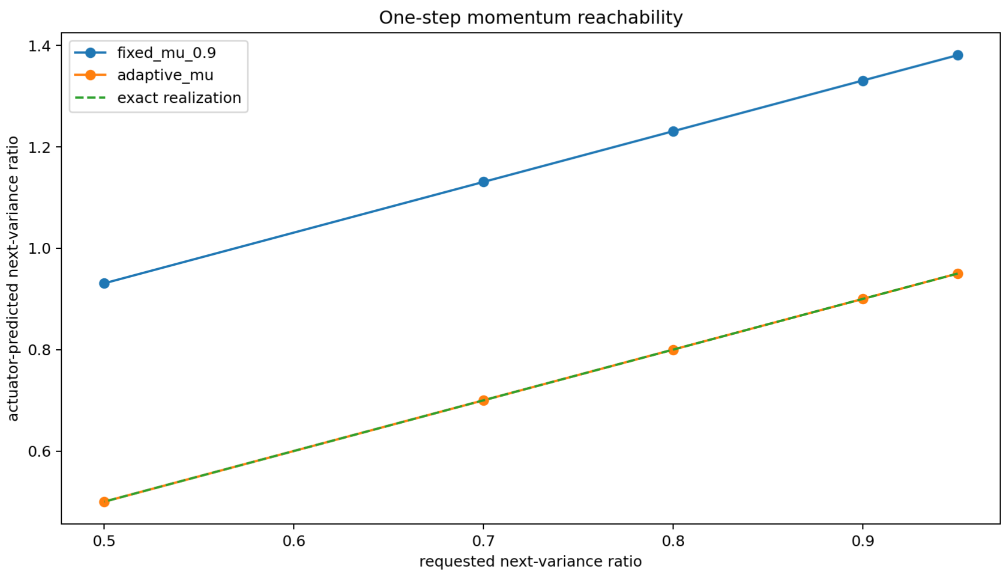


Figure 10.11: One-step reachability with fixed and adaptive momentum.

Chapter 11

Guarantees, limitations, and interpretation

11.1 Exact statements in the implemented model

For each scalar or diagonal block with fixed reference $X_{k+1} = a_0 X_k + \sqrt{r_0}\xi_k$, strictly positive r_0, Gaussian endpoint marginals, and finite horizon, the following statements are exact:

1. the endpoint coupling in Equation (4.4);
2. the covariance grid in Equation (4.8);
3. the adjacent cross-covariance in Equation (4.11);
4. the Markov bridge transition in Equation (4.12);
5. the Brownian formulas as the random-walk limit;
6. the one-step reachable set under the local scalar actuator model.

When an actuator exactly realizes the requested transition and the observation equals the true state, the next covariance equals the bridge covariance by construction.

11.2 What receding-horizon replanning means

Replanning does not mean that the realized closed loop is the original open-loop bridge solved at time zero. At each control opportunity, it solves a new finite-horizon bridge from the observed state to the same terminal intent relative to the current declared reference. The resulting policy is a model-predictive covariance controller.

The reference-consistency claim is narrower and important: within each replan, the marginal path and first transition come from one reference. This fixes the deepest internal inconsistency of the first design without claiming that a drifting adaptive plant has one immutable global bridge law.

11.3 Conditional closed-loop stability

Because observer errors depend on the controlled trajectory, a bound that treats transition error as exogenous is insufficient. The small-gain condition Equation (5.26) supplies a conditional local statement when the combined observer/controller map is Lipschitz in tracking error. The library does not estimate a universal L or certify global adaptive stability.

Practical consequences are:

- confidence and model adequacy must enter actuation;

- one-step calibration must be monitored online;
- poorly calibrated channels should return to shadow or baseline mode;
- bounded probing may be needed to retain identifiability;
- closed-loop adversarial simulation is part of testing, not optional decoration.

11.4 Momentum and optimizer memory

A low structural-adequacy score is a statement that the implemented plant model is unsupported. The PyTorch gate prevents silent active use but does not solve the augmented momentum bridge. A complete treatment would define

$$y_k = \begin{pmatrix} z_k \\ v_k \end{pmatrix}, \qquad y_{k+1} = \mathcal{A}_k y_k + \epsilon_k, \qquad \Sigma_{k+1} = \mathcal{A}_k \Sigma_k \mathcal{A}_k^\top + \mathcal{R}_k, \tag{11.1}$$

then solve a matrix-valued finite-horizon bridge. Until that exists, active Adam or momentum deployment is justified only where the observed block dynamics are adequately first order over the control interval.

11.5 Curvature endogeneity

The linear response Equation (5.15) captures only the first local derivative of sharpness with respect to step size; edge-of-stability behavior can be substantially more nonlinear [19]. Edge-of-stability training can involve delayed, nonlocal, or discontinuous adaptation. The trust-region cap and horizon pause are safeguards, not a proof that the response model remains valid.

A stronger implementation would identify $a(u)$ directly rather than infer it through a scalar curvature, or use a learned uncertainty-calibrated actuator model constrained by stability and positive innovation.

11.6 Centering and observability

The controlled width is defined relative to a center filter. Therefore there is no center-free scalar active variance. Changing the center timescale changes the plant observed by the controller. Any scientific comparison must keep the center convention fixed or treat it as an ablation.

Cold regimes pose an observability tradeoff familiar from adaptive and dual control [20]. Cooling suppresses the stochastic excitation needed to estimate contraction, while high memory makes prediction errors persist. Probing can improve identification but also changes the process. BridgeLearn records this expenditure; it does not solve the globally optimal dual-control problem.

11.7 Persistent excitation and dual-control limits

Antithetic probing repairs a specific order-2 information deficit. Its sandbox advantage assumes that the signed command is known, applied exactly once, and removed from the regression response. Clipping, saturation, delayed application, or coordinate mismatch can invalidate that accounting.

The policy is bounded and resource-accounted but is not the solution of a globally optimal dual-control problem. It probes only when a named contrast is uncertain, width is cold, stability permits it, and cumulative budget remains. A task deployment must measure probe harm as well as identification gain.

11.8 Heavy tails and robust widths

IQR, MAD, and trimmed scales reduce jump sensitivity but lose the exact covariance interpretation. Heavy-tailed stochastic-gradient evidence makes this more than a pathological corner case [21]. The finite-horizon Gaussian bridge then acts as a control template over a robust scale, not an exact path-space statement about the full distribution. This may still be useful, but reports should distinguish "Gaussian covariance steering" from "robust scale steering."

For strongly multimodal or state-dependent noise, mixture, particle, or transport-map representations may be necessary.

11.9 Covariance and generalization

Marginal variance is not sufficient to characterize implicit regularization. Even the full local Gaussian transition omits higher-order and global effects. Two matched-width controls may differ in learning rate, anisotropic update noise, temporal correlation, basin transitions, and weight-averaging behavior.

Therefore the framework makes no theorem that lower tracking error implies better generalization. Claims about large-learning-rate benefits should include matched-width branch comparisons and measurements such as:

- Hessian or sharpness proxies;
- SWA-versus-current-iterate gap;
- gradient-noise anisotropy;
- basin occupation and transition rates in controlled landscapes;
- representation or policy-space displacement;
- CAROM itinerary and causal trajectory metrics.

If these differ strongly under matched covariance paths, the endpoint objective must be enriched beyond active width.

11.10 Endpoint and reference selection remain modeling problems

Version 0.3 clarifies the knobs but does not eliminate them. A channel still needs a terminal width, a reference transition, and a nominal horizon. These should be derived where possible from:

- hardware and actuator bounds;
- a calibrated reachable floor;
- validation perturbation tolerance;
- a declared conservative baseline command;
- a task-specific terminal mode;
- a compute budget.

They should not be advertised as automatically easier than selecting a scheduler until an experiment demonstrates transfer or derivation across tasks.

11.11 Compute comparability

Example-equivalent sample count is more informative than update count when batch changes, but it is still not identical to wall clock or energy. Accumulation can have nonlinear hardware efficiency; recurrent sweeps and rollouts have different costs. Serious comparisons should report FLOP estimates or measured wall clock in addition to the library's abstract compute units.

11.12 Interpretation for CAROM

The CAROM 12k failure is strong motivation, not proof. The best checkpoint occurred before the stretched OneCycle peak, later annealing did not recover it, and edge prediction survived while execution dynamics deteriorated. These facts motivate blockwise guarded control and dense monitoring. They do not establish that a reference-consistent bridge controller would prevent the collapse.

The first decisive CAROM result would require multiple seeds, schedule-duration controls, calibrated one-step predictions, matched compute, and trajectory-aware outcomes. A useful negative result is also possible: BridgeLearn may accurately predict and control local widths while failing to improve endpoint or itinerary behavior. That would show that the chosen observable or endpoint objective is inadequate.

11.13 Software maturity

The library is alpha research software. Its strongest claims are formula correctness, tested local behavior, explicit diagnostics, and safer failure modes. It has not yet been integrated with the CAROM repository, a production predictive-coding codebase, or a backpropagation RL benchmark. Full matrix covariance steering, augmented momentum, automatic endpoint inference, distributed resource coordination, and formal dual control remain open.

Chapter 12

Experimental program

12.1 A staged evaluation strategy

A credible evaluation should separate model adequacy, transition control, task utility, and compute.

Stage 0: shadow identification. Run the baseline unchanged. Fit widths, centers, AR(1)/AR(2) models, curvature response, noise, and one-step uncertainty. Reject unsupported blocks.

Stage 1: local closed-loop tests. Use controlled quadratics, delayed observations, heavy tails, integer resources, momentum plants, and abrupt parameter changes. Require calibration and bounded control effort.

Stage 2: reference ablation. Compare the legacy Brownian-path/current-projector composition against a fixed-reference and a filtered-commanded-reference bridge. Measure hysteresis after saturation and recovery from identical states.

Stage 3: task controls. Enable one adequate block at fixed resource. Report task outcomes and mechanism metrics, not only tracking error.

Stage 4: shared resources. Add quantized compute-priced accumulation, rollout, or solver budgets. Match total examples/FLOPs and wall clock.

Stage 5: full architecture. Run multi-block, guarded, multi-seed studies with preregistered endpoints and stopping rules.

12.2 Core ablations implied by the critique

12.2.1 Reference consistency

Hold endpoint intent and actuator bounds fixed. Compare:

1. Brownian marginal path plus current-command KL projection;
2. Brownian path plus fixed-reference projection;
3. exact linear-Gaussian reference bridge;
4. exact bridge with slowly filtered commanded reference.

After an induced early saturation, return all systems to the same measured state and compare subsequent commands. Persistent branch dependence in the current-command condition is direct evidence of hysteresis.

12.2.2 Matched covariance, different temporal coupling

Construct two feasible controls with the same next variance but different roots or cross-time correlations. Compare:

- lag-one correlation and mixing;
- sharpness and Hessian proxies;
- SWA-versus-iterate gap;
- basin occupation in a double well;
- downstream task utility.

This tests whether active width is too coarse to represent the benefit attributed to large learning rates.

12.2.3 Cold identifiability and probing

Compare no probe, fixed small dither, and confidence-triggered bounded probing. Outcomes are observer bias, contraction/noise uncertainty, terminal tracking, task utility, probe expenditure, and overshoot. A successful probe policy should improve calibrated identification more than it harms the terminal objective.

12.2.4 Momentum and edge of stability

Use a known second-order momentum plant and a neural block under SGD with momentum or AdamW. Compare:

1. naive AR(1) control;
2. AR(1) control gated by adequacy;
3. an augmented-state controller when available;
4. trust-region-only fallback near edge of stability.

Measure lag-two residuals, command oscillation, calibration, bursts, and task outcomes.

12.2.5 Robust width

Inject controlled heavy-tail contamination or use naturally heavy-tailed gradient blocks. Compare variance, trimmed variance, IQR, and MAD planning observables. Report sensitivity to rare jumps and whether a robust controller sacrifices responsiveness to genuine regime changes.

12.3 Preregistered CAROM study

The CAROM evidence motivates a focused design rather than an unrestricted scheduler search. The 12k run peaked at update 3,000, deteriorated near the stretched learning-rate peak, and preserved edge prediction while routing/execution metrics collapsed [4]. The mechanism screen also showed incomplete fixed-point settling and high GLV integration cost [3].

12.3.1 Arms

Across at least three to five seeds:

1. original 12k OneCycle protocol;
2. lower-peak or early-decay conventional control;
3. BridgeLearn shadow mode over the original baseline;
4. active blockwise fixed-batch BridgeLearn;
5. active blockwise compute-priced quantized-accumulation BridgeLearn;
6. reference ablation: legacy Brownian composition versus reference-consistent planner.

A resume-from-4k-checkpoint low-learning-rate arm can separately test basin preservation.

12.3.2 Blocks and adequacy gates

Use at least routing/edge, operator core, GLV control, and execution/readout groups. Because CAROM uses AdamW, active control requires gray-box/black-box structural adequacy, calibrated sharpness funnels, and acceptable companion conditioning per group. Inadequate groups remain at baseline, use trust-region-only movement, or move to a richer augmented-state adapter.

12.3.3 Outcomes

Primary task outcomes:

- fixed-corpus L2–4 endpoint accuracy;
- held-out L5 accuracy;
- natural-minus-shuffled causal gap.

Mechanism outcomes:

- edge accuracy and per-loss terms;
- itinerary order, coverage, exact order, and transition recall;
- routing entropy, GLV dwell, activity, and halt statistics;
- gradient/update norms and blockwise sharpness proxies.

Controller outcomes:

- predicted–realized one-step calibration per block;
- reference drift and structural adequacy;
- step/resource saturation, probes, and horizon extension;
- examples, GLV steps, wall clock, and progress per compute.

Checkpoint densely through updates 2,000–5,000. Predeclare a fixed-panel guard combining endpoint and trajectory metrics. Selection rules must be fixed before inspecting outcomes.

12.4 Predictive-coding and fixed-point experiments

12.4.1 Adaptive inference depth

On a predictive-coding or equilibrium task, compare fixed sweeps with an episodic reference bridge. Match solver evaluations or wall clock. Require both terminal bridge progress and an absolute residual/energy criterion. Measure accuracy, residual, energy monotonicity, calibration, and sweep distribution by input difficulty.

12.4.2 Dual probing in latent inference

In a cold residual regime, evaluate whether bounded state-noise probes improve contraction identification and stopping calibration. Because probing changes inference, measure final reconstruction/task loss and repeatability, not only observer variance.

12.4.3 Persistent local learning

Use separate episodic activity and persistent synaptic channels. Compare coupled versus independent progress and test whether freezing weight progress on failed inference episodes improves stability.

12.5 Backpropagation reinforcement learning

Natural channels are actor, critic, encoder, entropy temperature, and target-network update. The actor should retain an independent policy-space trust region even if local curvature inversion permits a larger step. The critic reference should be invalidated after major target or data-distribution changes.

Report return distributions, environment interactions, policy KL, entropy, critic calibration, constraint violations, sample efficiency, and wall clock. Compute matching must include rollout collection; a variable minibatch controller should not receive free data.

12.6 Falsification criteria

The framework is weakened if any of the following recur across well-powered studies:

- exact reference-consistent transitions are systematically uncalibrated despite adequate local diagnostics;
- horizon governance spends substantially more compute without task benefit;
- matched-width controls show task differences entirely unrelated to measured transition or richer observables;
- structural adequacy is low for nearly every practical block, making the scalar controller inactive;
- endpoint and reference selection require as much task-specific search as conventional schedules with no transfer;
- robust widths or probes improve tracking but reliably degrade task outcomes;
- blockwise control fails to localize the CAROM execution collapse.

A negative result may motivate a richer state, observable, or utility objective rather than another cosmetic schedule curve.

12.7 Reporting standard

Every experimental table should place four categories side by side:

1. **task**: accuracy, return, residual, generalization, or causal metrics;

2. **model**: one-step calibration, adequacy, center and robust-scale diagnostics;
3. **control**: tracking, saturation, horizon, probe, and command variation;
4. **compute**: examples, rollouts, sweeps, FLOPs or wall clock.

Tracking RMSE should never appear alone as the headline result. Code, controller checkpoints, traces, calibration artifacts, and fixed evaluation panels should be archived with exact version and checksum information.

Chapter 13

Conclusion

Brownian Schrodinger bridges provide a deeper mathematical explanation for OneCycle-like schedules: covariance projections are quadratic in intrinsic time, endpoint asymmetry determines temporal skew, and Gaussian endpoints yield closed-form peak and phase-transition laws. This explains why a warm-up/hump/anneal family can arise from boundary conditions rather than from hand-drawn curves.

Turning that explanation into a learning algorithm required several separations. Marginal spread is not local diffusion. Consecutive marginal covariances do not determine temporal coupling. A bridge is defined relative to one reference process, so path and transition must share that reference. Learning rate affects both deterministic memory and stochastic injection, whereas batch or explicit noise can act more selectively. Physical update count is not intrinsic bridge time, and infeasibility must be paid through extra horizon, changed momentum, additional compute, or relaxed intent.

Reference-consistent Bridge Learning encodes these separations as a receding-horizon covariance-steering framework. A declared or identified linear-Gaussian reference generates a coherent finite-horizon sequence of marginals, adjacent cross-covariances, contractions, and innovations. A controller observes the local plant, tests the requested transition against trust, resources, and compute, realizes what is feasible, records the prediction, and replans from the realized state.

Version 0.3 extends the framework where realistic large-step training first breaks a first-order model. Sharpness is promoted from a static response curve to a slow identified state with progressive sharpening, edge restoration, robust funnels, an adequacy ladder, and jump detection. Momentum is promoted from a hidden optimizer detail to a companion-form plant with gray-box identification, order-2 antithetic excitation, Jury stability geometry, Lyapunov transient diagnostics, and an explicit cooling-reachability floor. A shared dominant-sharpness guard acknowledges that the globally binding Hessian direction can migrate across blocks.

The sandbox suite supports these mechanisms within their declared models. Dynamic sharpness prediction outperformed a rolling static response; antithetic probing dramatically improved the weak AR(2) contrast per unit width disturbance; a stable near-repeated-root companion matrix produced large transient covariance amplification that spectral radius missed; and fixed high momentum imposed a minimum cooling horizon. These are testable control components, not task-level proofs.

The software architecture follows the same epistemic discipline. `bridgelearn` controls transitions rather than gradients, supports persistent and episodic channels, exposes model adequacy and failure states, prices discrete resources, checkpoints unfinished probe pairs, and treats one-step calibration as a primary artifact. Backpropagation, predictive coding, fixed-point inference, reinforcement learning, and CAROM enter through observers and actuators rather than separate mathematical cores.

The framework's strongest present claim is:

> Boundary conditions and a declared stochastic reference can define a coherent desired transition law. A low-dimensional identified plant can determine whether and how an iterative learning system may realize that law, while making uncertainty, momentum,

probing, physical time, and compute explicit.

The decisive evidence must now come from compute-matched, multi-seed task experiments. For CAROM, that means dense monitoring through the prior failure window, blockwise sharpness and structural adequacy, causal trajectory metrics, and a preregistered validation guard. For predictive coding, it means adaptive inference depth with absolute residual criteria and calibrated episodic transitions. For reinforcement learning, it means actor/critic separation, rollout-normalized compute, and policy-space safeguards. A locally calibrated bridge controller may still be globally irrelevant; that negative result would identify a missing state, observable, or endpoint utility rather than motivate another cosmetic schedule curve.

Chapter 14

Derivations and proofs

14.1 Quadratic covariance law for Brownian bridges

Let the reference process on $[0, T]$ be

$$dX_t = \sqrt{\varepsilon}\, dW_t, \tag{14.1}$$

and write $s = t/T$ and $q = \varepsilon T$. A Schrodinger bridge relative to this reference has density with respect to the reference of the endpoint-factor form

$$\frac{dP^\star}{dR} = f(X_0)g(X_T). \tag{14.2}$$

Consequently, conditioning on (X_0, X_T) cancels the endpoint factors, so the bridge and the reference have the same conditional path law.

For Brownian motion,

$$X_s \mid X_0, X_T \sim \mathcal{N}\left((1-s)X_0 + sX_T,\ qs(1-s)I\right). \tag{14.3}$$

Equivalently,

$$X_s = (1-s)X_0 + sX_T + \sqrt{qs(1-s)}Z, \qquad Z \sim \mathcal{N}(0, I), \tag{14.4}$$

with Z independent of the endpoint pair under the bridge endpoint coupling.

Let

$$\Sigma_0 = \mathrm{Cov}(X_0), \qquad \Sigma_T = \mathrm{Cov}(X_T), \qquad \Gamma = \mathrm{Cov}(X_0, X_T). \tag{14.5}$$

The law of total covariance gives

$$\begin{aligned} \Sigma_s &= \mathrm{Cov}\big((1-s)X_0 + sX_T\big) + qs(1-s)I & (14.6)\\ &= (1-s)^2\Sigma_0 + s^2\Sigma_T + s(1-s)(\Gamma + \Gamma^\top + qI). & (14.7) \end{aligned}$$

This proves Theorem 3.1. No Gaussian assumption on the endpoint marginals is required; finite second moments are sufficient.

For a fixed positive-semidefinite metric M, the scalarization

$$V_M(s) = \mathrm{tr}(M\Sigma_s) \tag{14.8}$$

is also quadratic. Thus every frozen metric projection has at most one interior extremum.

14.2 Peak location for a general scalar projection

Write

$$V(s) = A(1-s)^2 + Bs^2 + Ks(1-s). \tag{14.9}$$

Expanding gives

$$V(s) = A + (K-2A)s + (A+B-K)s^2. \tag{14.10}$$

The curve is strictly concave when

$$D = K - A - B > 0. \tag{14.11}$$

Its stationary point is

$$s_\star = \frac{K-2A}{2D} = \frac{1}{2} + \frac{B-A}{2D}. \tag{14.12}$$

It lies in $(0, 1)$ precisely when

$$K > 2A \qquad \text{and} \qquad K > 2B. \tag{14.13}$$

Therefore, whenever an interior maximum exists,

$$\operatorname{sign}\left(s_\star - \frac{1}{2}\right) = \operatorname{sign}(B-A). \tag{14.14}$$

The endpoint coupling changes D and hence the magnitude of the shift, but not its direction.

14.3 Gaussian endpoint coupling

Let scalar endpoint marginals have variances A and B, and let $C = \operatorname{Cov}(X_0, X_T)$. Among Gaussian couplings, the endpoint Schrodinger objective differs from a constant by

$$J(C) = -\frac{C}{q} - \frac{1}{2}\log\left(1 - \frac{C^2}{AB}\right), \qquad |C| < \sqrt{AB}. \tag{14.15}$$

The first term is the cross term from Brownian transport cost; the second is the mutual-information contribution of the coupling. Differentiating,

$$J'(C) = -\frac{1}{q} + \frac{C}{AB - C^2}. \tag{14.16}$$

The optimum solves

$$C^2 + qC - AB = 0. \tag{14.17}$$

The admissible root is

$$C_q = \frac{\sqrt{q^2 + 4AB} - q}{2}. \tag{14.18}$$

Let

$$S = \sqrt{q^2 + 4AB}. \tag{14.19}$$

Since $2C_q + q = S$, the bridge variance is

$$V(s) = A(1-s)^2 + Bs^2 + Ss(1-s). \tag{14.20}$$

14.4 Gaussian peak height and phase boundaries

Define

$$D = S - A - B. \tag{14.21}$$

In the interior-hump regime $S > 2\max(A, B)$,

$$s_\star = \frac{S - 2A}{2D} = \frac{1}{2} + \frac{B - A}{2D}. \tag{14.22}$$

Writing the quadratic in vertex form gives

$$V_{\max} = A + \frac{(S - 2A)^2}{4D}. \tag{14.23}$$

The numerator simplifies because

$$4AD + (S - 2A)^2 = 4A(S - A - B) + S^2 - 4AS + 4A^2 \tag{14.24}$$

$$= S^2 - 4AB = q^2. \tag{14.25}$$

Therefore

$$V_{\max} = \frac{q^2}{4D}. \tag{14.26}$$

The terminal standard-deviation floor relative to the maximum is

$$\frac{\sqrt{B}}{\sqrt{V_{\max}}} = \frac{2\sqrt{BD}}{q}. \tag{14.27}$$

The initial boundary becomes the maximum when the initial derivative is nonpositive:

$$S \le 2A. \tag{14.28}$$

Equality gives

$$B_- = A - \frac{q^2}{4A}. \tag{14.29}$$

When $B_- \le 0$, no positive terminal variance produces this start-pinned phase. The terminal boundary becomes the maximum when

$$S \le 2B, \tag{14.30}$$

with equality at

$$B_+ = \frac{A + \sqrt{A^2 + q^2}}{2}. \tag{14.31}$$

Thus the scalar Gaussian family passes from cooling-only, through an interior one-cycle hump, to warming-only as the terminal width increases.

14.5 Time change and intrinsic progress

For a reference diffusion

$$\mathrm{d}X_t = \sqrt{a(t)}\,\mathrm{d}W_t, \tag{14.32}$$

define cumulative quadratic variation

$$Q_t = \int_0^t a(u)\,\mathrm{d}u, \qquad s(t) = \frac{Q_t}{Q_T}. \tag{14.33}$$

The endpoint-conditioned bridge has mean interpolation weights $1 - s(t)$ and $s(t)$ and conditional covariance

$$Q_T s(t)(1 - s(t))I. \tag{14.34}$$

Hence all preceding covariance formulas hold after replacing normalized physical time by $s(t)$. Endpoint marginals and total quadratic variation determine the path in intrinsic time but do not identify $a(t)$ or the physical-time peak. This is the mathematical basis for a separate bridge clock.

14.6 Exact finite-horizon linear-Gaussian bridge

Consider the scalar reference

$$X_{k+1} = a_0 X_k + \sqrt{r_0}\,\xi_k, \qquad k = 0, \dots, N-1.$$

Assume the reference initial marginal already equals $\mu_0 = \mathcal{N}(0, V_0)$ and impose terminal marginal $\mu_N = \mathcal{N}(0, V_N)$. Write

$$\Phi = a_0^N, \qquad Q = r_0 \sum_{j=0}^{N-1} a_0^{2j}.$$

Then the reference endpoint kernel is $X_N \mid X_0 = x \sim \mathcal{N}(\Phi x, Q)$.

14.6.1 Endpoint coupling

Any zero-mean Gaussian coupling of the endpoint marginals has covariance

$$\Sigma_{0N} = \begin{pmatrix} V_0 & C \\ C & V_N \end{pmatrix}, \qquad C^2 < V_0 V_N.$$

Under this coupling,

$$X_N \mid X_0 = x \sim \mathcal{N}\left(\frac{C}{V_0} x, V_N - \frac{C^2}{V_0}\right).$$

The conditional relative entropy to the reference endpoint kernel, averaged over X_0, is

$$J(C) = \frac{1}{2}\left[\log \frac{Q}{V_N - C^2/V_0} + \frac{V_N - 2\Phi C + \Phi^2 V_0}{Q} - 1\right]. \tag{14.35}$$

Differentiation gives

$$J'(C) = \frac{C}{V_0 V_N - C^2} - \frac{\Phi}{Q}.$$

The minimizing cross-covariance therefore solves

$$\Phi C^2 + QC - \Phi V_0 V_N = 0. \tag{14.36}$$

The admissible root with the sign of Φ is

$$C = \frac{-Q + \sqrt{Q^2 + 4\Phi^2 V_0 V_N}}{2\Phi} = \frac{2\Phi V_0 V_N}{Q + \sqrt{Q^2 + 4\Phi^2 V_0 V_N}}, \tag{14.37}$$

where the second form remains stable as $\Phi \to 0$.

14.6.2 Reference bridge conditional on endpoints

For $0 \le k \le N$, write

$$X_k = \phi_k X_0 + \eta_k, \qquad \phi_k = a_0^k, \qquad \mathrm{Var}(\eta_k) = Q_k = r_0 \sum_{j=0}^{k-1} a_0^{2j}.$$

Let $b_k = a_0^{N-k}$. Since $\eta_N = b_k \eta_k +$ independent future noise,

$$\mathrm{Cov}(\eta_k, \eta_N) = b_k Q_k.$$

Conditioning the jointly Gaussian noise pair on $\eta_N = X_N - \Phi X_0$ gives

$$\begin{aligned} \mathbb{E}[X_k \mid X_0, X_N] &= \phi_k X_0 + \frac{b_k Q_k}{Q}(X_N - \Phi X_0) && (14.38)\\ &= \alpha_k X_0 + \beta_k X_N, && (14.39) \end{aligned}$$

where α_k and β_k are defined in Equation (4.5). The conditional variance is

$$S_k = Q_k - \frac{(b_k Q_k)^2}{Q}.$$

Averaging over the optimal endpoint coupling yields Equation (4.8).

14.6.3 Adjacent cross-covariance

Before endpoint conditioning,

$$\mathrm{Cov}(\eta_k, \eta_{k+1}) = a_0 Q_k.$$

Gaussian conditioning on η_N subtracts the rank-one term

$$\frac{\mathrm{Cov}(\eta_k, \eta_N)\,\mathrm{Cov}(\eta_{k+1}, \eta_N)}{Q},$$

which gives Equation (4.9). Adding the endpoint terms from $\alpha_k X_0 + \beta_k X_N$ produces Equation (4.11).

Finally, for a jointly Gaussian pair (X_k, X_{k+1}), linear regression gives

$$\mathbb{E}[X_{k+1} \mid X_k] = \frac{K_{k,k+1}}{V_k} X_k.$$

The conditional variance is the Schur complement

$$V_{k+1} - \frac{K_{k,k+1}^2}{V_k} \ge 0,$$

proving Equation (4.12).

14.7 Legacy one-step Gaussian KL projection

Let the current centered scalar state have variance V. The reference transition is

$$z^+ = a_0 z + \epsilon_0, \qquad \epsilon_0 \sim \mathcal{N}(0, r_0), \tag{14.40}$$

and a candidate transition is

$$z^+ = az + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, r). \tag{14.41}$$

The target next variance is W, so

$$r = W - a^2 V > 0. \tag{14.42}$$

Averaging the conditional Gaussian relative entropy over $z \sim \mathcal{N}(0, V)$ yields

$$J(a, r) = \frac{1}{2}\left[\log \frac{r_0}{r} + \frac{r + (a - a_0)^2 V}{r_0} - 1\right]. \tag{14.43}$$

After substituting $r = W - a^2 V$, differentiation gives

$$\frac{\mathrm{d}J}{\mathrm{d}a} = V\left(\frac{a}{W - a^2 V} - \frac{a_0}{r_0}\right). \tag{14.44}$$

The function $a/(W - a^2 V)$ is strictly increasing on the feasible interval, so the stationary point is unique. It solves

$$a_0 V a^2 + r_0 a - a_0 W = 0. \tag{14.45}$$

For $a_0 \neq 0$, the root continuous with the reference sign is

$$\widetilde{a} = \frac{2 a_0 W}{r_0 + \sqrt{r_0^2 + 4 a_0^2 V W}}, \tag{14.46}$$

with

$$\widetilde{r} = W - \widetilde{a}^2 V. \tag{14.47}$$

For $a_0 = 0$, the minimizer is $\widetilde{a} = 0$ and $\widetilde{r} = W$. Bounds on contraction or innovation are handled by minimizing the same one-dimensional objective on the feasible interval.

14.8 Learning-rate reachability and exact roots

In the scalar local SGD model,

$$W(\eta) = (1 - \eta h)^2 V + \eta^2 q, \qquad q = \frac{\nu}{m}. \tag{14.48}$$

Let

$$H = h^2 V + q. \tag{14.49}$$

Completing the square gives

$$W(\eta) = \frac{qV}{H} + H\left(\eta - \frac{hV}{H}\right)^2. \tag{14.50}$$

Therefore the one-step reachability condition is

$$W \geq W_{\min} = \frac{qV}{h^2 V + q}. \tag{14.51}$$

When feasible, the two exact roots are

$$\eta_{\pm} = \frac{hV \pm \sqrt{(h^2 V + q) W - qV}}{h^2 V + q}. \tag{14.52}$$

At $W = V$, the roots reduce to

$$\eta_{-} = 0, \qquad \eta_{+} = \frac{2hV}{h^2 V + q}. \tag{14.53}$$

They preserve the same marginal variance but have different lag-one correlation and therefore different temporal coupling. The discriminant sensitivity

$$\frac{\partial \eta_{\pm}}{\partial W} = \pm \frac{1}{2\sqrt{(h^2 V + q) W - qV}} \tag{14.54}$$

diverges at the reachability boundary, explaining why exact root chasing is ill-conditioned.

14.9 Equilibrium map and adiabatic tracking

For fixed η, h, and q, the stationary variance satisfies

$$V_\infty = \frac{\eta q}{h(2-\eta h)}, \qquad 0 < \eta h < 2. \tag{14.55}$$

Inverting gives

$$\eta_{\mathrm{eq}}(V) = \frac{2hV}{q + h^2 V}. \tag{14.56}$$

Suppose this map is evaluated at a moving target $V_k^\star$, and let

$$e_k = V_k - V_k^\star. \tag{14.57}$$

Then

$$e_{k+1} = \alpha_k e_k + V_k^\star - V_{k+1}^\star, \tag{14.58}$$

where

$$\alpha_k = \left(\frac{q_k - h_k^2 V_k^\star}{q_k + h_k^2 V_k^\star}\right)^2. \tag{14.59}$$

If $\alpha_k \le \rho < 1$, then

$$|e_k| \le \rho^k |e_0| + \sum_{j=0}^{k-1} \rho^{k-1-j} |V_j^\star - V_{j+1}^\star|. \tag{14.60}$$

This is an adiabatic tracking bound: slowly varying targets are followed accurately, but cold low-width regimes may have α near one and require a slower clock.

14.10 Cross-time covariance and unique Gaussian realization

Let consecutive zero-mean Gaussian states have joint covariance

$$\begin{bmatrix} \Sigma & C^\top \\ C & \Sigma_+ \end{bmatrix} \succeq 0, \tag{14.61}$$

where $C = \mathrm{Cov}(z_+, z)$. Gaussian conditioning yields

$$z_+ = Az + \epsilon, \qquad A = C\Sigma^{-1}, \tag{14.62}$$

with

$$R = \mathrm{Cov}(\epsilon) = \Sigma_+ - C\Sigma^{-1}C^\top \succeq 0. \tag{14.63}$$

The innovation covariance is the Schur complement of Σ. Thus the pair of marginal covariances does not determine a dynamics, but adding cross-time covariance determines the Gaussian Markov transition uniquely.

This is the precise mathematical content of the movement/temperature decomposition:

$$C\Sigma^{-1} \longleftrightarrow \text{memory and contraction}, \qquad \Sigma_+ - C\Sigma^{-1}C^\top \longleftrightarrow \text{new innovation}. \tag{14.64}$$

14.11 Recursive identification of the sharpness plant

From Equation (6.9), define the forgetting-factor RLS recursion

$$K_k = \frac{P_{k-1}\varphi_k}{\lambda + \varphi_k^\top P_{k-1}\varphi_k}, \tag{14.65}$$

$$\widehat{\beta}_k = \widehat{\beta}_{k-1} + K_k(y_k - \varphi_k^\top \widehat{\beta}_{k-1}), \tag{14.66}$$

$$P_k = \lambda^{-1}(I - K_k\varphi_k^\top)P_{k-1}. \tag{14.67}$$

Projection onto nonnegative rates can be applied after each update. Residual quantiles and parameter covariance induce a one-step interval, which is rolled forward by interval arithmetic. Because the restoration feature depends monotonically on positive edge excess for fixed commands, conservative lower and upper recursions remain scalar.

14.12 Jury triangle and the momentum edge

For $p(s) = s^2 - c_1 s - c_2$, the second-order Jury test gives

$$1 - c_1 - c_2 > 0, \quad 1 + c_1 - c_2 > 0, \quad 1 + c_2 > 0.$$

Substitute $c_1 = 1 + \mu - \eta h$ and $c_2 = -\mu$. The first inequality becomes $\eta h > 0$; the second becomes $2(1+\mu) - \eta h > 0$; the third gives $1 - \mu > 0$. Together with $\mu \geq 0$, this yields Equation (6.4). The discriminant boundary $c_1^2 + 4c_2 = 0$ separates real and complex roots. In the complex case the root product is $-c_2 = \mu$, hence radius $\sqrt{\mu}$.

14.13 Gray-box momentum regression

Starting from Equation (6.1), move known command terms to the left and divide by η_k to obtain Equation (6.12). With weights proportional to m_k, weighted least squares estimates h while accounting for heteroscedastic variance v/m_k. If z_k is measured with additive error, ordinary least squares attenuates the slope. Deeper lags can act as instruments when they correlate with latent z_k but are independent of current measurement error.

14.14 Why white amplitude does not repair the cold AR(2) contrast

For stationary regressors (z_k, z_{k-1}), the moment matrix has the form

$$M = \gamma_0 \begin{bmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{bmatrix}. \tag{14.68}$$

Its eigenvectors are $(1,1)$ and $(1,-1)$ with eigenvalues $\gamma_0(1+\rho_1)$ and $\gamma_0(1-\rho_1)$. In a cold overdamped regime $\rho_1 \to 1$, so the contrast direction is starved. White driving scales state variance and innovation together; it does not relocate spectral mass toward the missing temporal direction. Alternating excitation has frequency π and regressor direction $(1,-1)$, directly increasing the weak eigenvalue.

14.15 Lyapunov-adapted covariance contraction

Let $P \succ 0$ satisfy Equation (6.15). Then

$$\begin{aligned} \|AEA^\top\|_P &= \|P^{1/2}AEA^\top P^{1/2}\|_F && (14.69)\\ &\le \|P^{1/2}AP^{-1/2}\|_2^2\|E\|_P && (14.70)\\ &\le \bar{\rho}^2\|E\|_P. && (14.71) \end{aligned}$$

With disturbance gain L_P and additive bound $d_{\max}$,

$$\|E_{k+1}\|_P \le (\bar{\rho}^2 + L_P)\|E_k\|_P + d_{\max}.$$

When Equation (6.17) holds, the asymptotic P-norm bound is $d_{\max}/(1 - \bar{\rho}^2 - L_P)$. Conversion to Euclidean norm contributes at most $\sqrt{\kappa(P)}$, yielding Equation (6.18).

14.16 Momentum cooling horizon

In an underdamped fixed-momentum mode, state amplitude decays asymptotically no faster than $(\sqrt{\mu})^N$ and variance no faster than μ^N. To reach $V_N/V_0 \le r$, require $\mu^N \le r$, hence

$$N \ge \frac{\log r}{\log \mu}.$$

Taking the ceiling yields Equation (6.19). This is an optimistic lower bound because it omits innovation, non-normal transient growth, and command constraints.

Chapter 15

Software reference and reproducibility

15.1 Installation

From the source tree:

```
python -m pip install -e .
```

Listing 15.1: Editable installation.

From the wheel:

```
python -m pip install bridgelearn-0.3.0-py3-none-any.whl
```

Listing 15.2: Wheel installation.

The core dependency is NumPy. Optional extras install PyTorch, Matplotlib, or development tools.

15.2 Principal public objects

Table 15.1: Principal objects in `bridgelearn` v0.3.0.

Object	Role
`GaussianState`	Blockwise width, confidence, and labels.
`GaussianTransition`	Blockwise contraction and innovation.
`LocalDynamics`	Width, curvature, noise, current controls, response slope, trust cap, and structural adequacy.
`LinearGaussianBridgePath`	Exact finite-horizon scalar/diagonal bridge relative to an AR(1) reference.
`GaussianBridgePath`	Legacy Brownian Gaussian covariance path for theory and ablations.
`ReferenceBridgePlanner`	Receding-horizon exact bridge planner with horizon governance.
`FixedReferencePolicy`	Fixed declared reference transition.
`CallbackReferencePolicy`	Application-defined commanded reference.
`FilteredReferencePolicy`	Slowly filtered contraction and log innovation.
`ObservedReferencePolicy`	Current observed transition; compatibility/ablation mode.
`ConfidenceProbePolicy`	Bounded low-confidence innovation probe.
`StepBatchActuator`	Step/contraction plus quantized compute-priced shared batch.

Object	Role
`StepNoiseActuator`	Step/contraction plus explicit noise innovation.
`StepOnlyActuator`	Fixed-resource step-only fallback.
`DirectTransitionActuator`	Direct transition controls for simulators or special systems.
`DynamicsAdapter`	Callback-based dynamics observation and control application.
`CallbackAdapter`	Fully custom channel adapter.
`BridgeController`	Persistent named-channel orchestration and shadow mode.
`BridgeEpisode`	Isolated episodic controller.
`CenteredWidthObserver`	Explicit center filter plus scale estimator.
`VarianceScale`	Ordinary variance.
`TrimmedVarianceScale`	Tail-trimmed variance.
`IQRScale`	IQR-based robust variance-equivalent scale.
`MADScale`	MAD-based robust variance-equivalent scale.
`AR2AdequacyObserver`	structural adequacy under possible momentum/second-order dynamics.
`CurvatureResponseObserver`	First-order curvature response to step size.
`BridgeTrace`	JSON/CSV control and observation trace.
`CalibrationReport`	One-step bias, errors, correlation, quantiles, and coverage.
`TorchOptimizerAdapter`	PyTorch optimizer-group integration with momentum adequacy gate.

15.3 Momentum-aware objects

The v0.3 public API additionally exports the following groups of objects:

- dynamic sharpness: `SharpnessPlant`, `CurvatureFunnel`, and the curvature adequacy/model-ladder utilities;
- safety coordination: the jump detector and dominant-sharpness coordinator;
- momentum identification: the gray-box, forgetting-factor, and structural-adequacy observers;
- active control: `AntitheticProbePolicy`, the momentum step/batch actuator, and the sharpness-aware reference-bridge planner;
- companion utilities for Jury margins, roots, damping class, Lyapunov metrics, common-metric checks, and small-gain reports.

A checkpoint must preserve sharpness RLS state and residual history, curvature-model adequacy, gray-box and black-box identifiers, probe budget and unfinished antithetic pair, previous width and lag covariance, momentum commands, Lyapunov and slew diagnostics, planner horizon, reference policy, cumulative compute, and trace/calibration state.

15.4 Reference-consistent persistent example

The complete runnable script is `examples/reference_consistent_quadratic.py`. Its essential construction is:

```
reference = GaussianTransition(
    contraction=[0.995],
    innovation=[0.001],
```

```
)

planner = ReferenceBridgePlanner(
    terminal_variance=[0.005],
    total_steps=200,
    reference_policy=FixedReferencePolicy(reference),
    probe_policy=ConfidenceProbePolicy(
        confidence_threshold=0.45,
        cold_variance=0.012,
        max_probe_innovation=2.5e-5,
        max_total_probe=5e-4,
    ),
    max_horizon_multiplier=64.0,
)

actuator = StepBatchActuator(
    step_bounds=(1e-5, 0.20),
    batch_bounds=(16.0, 512.0),
    batch_quantum=16.0,
    compute_reference=64.0,
    compute_price=0.004,
    resource_deadband=0.06,
)
```

Listing 15.3: Core configuration of the reference-consistent example.

After each plant step, the example appends the command and realized width to a `BridgeTrace`, then writes JSON, CSV, a `CalibrationReport`, and plots.

15.5 PyTorch integration

The adapter expects one bridge block per optimizer parameter group. A statistics callback returns `TorchGroupStatistics`. For Adam or momentum, it must include `ar1_adequacy` unless the caller explicitly opts into a first-order approximation.

```python
def statistics(context):
    return TorchGroupStatistics(
        variance=context.block_widths,
        curvature=context.block_curvature,
        noise_scale=context.block_noise,
        effective_batch=context.effective_batch,
        confidence=context.block_confidence,
        curvature_slope=context.curvature_slope,
        trust_region_step=context.trust_steps,
        structural_adequacy=context.structural_adequacy,
        labels=context.block_names,
    )
```

Listing 15.4: Conceptual PyTorch statistics callback.

Gradient accumulation remains a training-loop responsibility because changing it changes forward/backward execution and compute. The adapter calls an application-supplied resource setter.

15.6 Episodic relaxation pattern

```python
with controller.episode("activities") as episode:
    while not episode.done:
        command = episode.plan(state)
        episode.apply(state, command)

        state.relax_once()

        episode.after_step(state, command)
```

Listing 15.5: Reference-consistent episodic relaxation loop.

The planner and its filtered-reference state are isolated in the episode copy.

15.7 Reproducing tests and examples

From the source repository:

```
PYTHONPATH=src pytest -q
python -m compileall -q src examples tests
PYTHONPATH=src python examples/reference_consistent_quadratic.py
PYTHONPATH=src python examples/predictive_relaxation.py \
    --output-dir artifacts/predictive_relaxation
```

Listing 15.6: Commands used to validate v0.3.0.

The test command reports thirty-eight passing tests. The examples produce machine-readable summaries and traces in addition to plots.

15.8 Calibration workflow

```python
trace.append(
    command,
    observed_variance=observed_width,
    observed_variance_std=standard_error,
)

report = trace.calibration_report(nominal_coverage=0.90)
report.to_json("calibration.json")
```

Listing 15.7: Writing a first-class calibration artifact.

Calibration should be evaluated before task-performance interpretation or active deployment.

15.9 CAROM integration materials

The software bundle includes:

- `docs/CAROM_INTEGRATION.md`, a critique-aware deployment plan;
- `examples/carom_skeleton.py`, a shadow-mode reference-consistent weight-channel skeleton;
- `docs/CRITIQUE_RESPONSE.md`, a point-by-point record of accepted changes and remaining limitations.

The skeleton uses a slowly filtered callback reference evaluated at declared baseline learning rates and batch, requires AdamW structural-adequacy diagnostics, prices and quantizes accumulation, enables bounded probing, and uses a trajectory-aware validation guard.

15.10 Recommended integration workflow

1. Run shadow mode and declare the controlled observable and center.
2. Verify robust versus ordinary width behavior.
3. Fit the commanded reference and test AR(1)/AR(2) adequacy.
4. Verify one-step calibration and uncertainty coverage.
5. Enable one adequate persistent block at fixed resource.
6. Add compute-priced resource or explicit-noise control.
7. Add bounded probing only after cold observability is demonstrated.
8. Add episodic stopping and multi-channel coordination.
9. Preserve complete controller state and calibration in checkpoints.

15.11 Source-level reproducibility of this report

The report is distributed as ASCII LaTeX together with figures, numerical summaries, traces, build notes, and checksums. Build with:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error bridge_learning_v0_3.tex
```

Listing 15.8: Building the revised report.

All non-ASCII mathematical symbols are generated by LaTeX commands.

Chapter 16

Notation and terminology

Table 16.1: Notation used throughout the report.

Symbol	Meaning
T or N	Physical or discrete bridge horizon.
$s \in [0, 1]$	Normalized intrinsic progress summary.
ε	Brownian reference diffusion coefficient.
$q = \varepsilon T$	Brownian quadratic-variation budget in the explanatory model.
a_0, r_0	Contraction and innovation of the declared scalar reference.
ϕ_k, Q_k	Reference transition factor and finite-horizon Gramian.
C_{0N}	Endpoint cross-covariance of the linear-Gaussian bridge.
V_k	Scalar active width of a controlled channel.
A_k, R_k	Matrix contraction and innovation covariance.
a_k, r_k	Scalar contraction and innovation variance.
h_k	Effective positive curvature or contraction coefficient.
η_k	Learning, relaxation, or integration step.
ν_k	Unit-resource stochastic update-noise scale.
m_k	Effective batch, accumulation, rollout, or related resource.
M_k	Metric used to scalarize covariance into active width.
V_T	Desired terminal active width.
p_k	Identification-probe innovation.
N_k	Planner remaining horizon.
C_k	Accumulated physical compute proxy.
$\mathrm{KL}(P\|R)$	Relative entropy of path measure P with respect to reference R.
`ReferencePolicy`	Object selecting the declared local reference transition.
`ReferenceBridgePlanner`	Receding finite-horizon bridge and horizon governor.
`Actuator`	Object projecting target transitions onto feasible controls.
`Adapter`	System-specific observation and control interface.
`CalibrationReport`	Predicted-versus-realized one-step diagnostic artifact.

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